# The Trading and Quant Research Map

A triaged survey of what is worth learning,
what is tradeable, and what will waste your time

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Built from fifteen independent research agents across two rounds, plus one adversarial completeness critic. Every source was filtered mechanically: it qualifies only if at least two of five practitioner tests hold, and any backtest presented without a cost model was rejected outright regardless of what else it passed. Claims a second round could not verify are marked as such rather than smoothed over. Where the two rounds disagreed, the disagreement was adjudicated against primary data and the adjudication is shown.

This document reports what is documented. It contains no investment advice, no recommended allocations and no trade suggestions. Tax and regulatory material states rules and their arithmetic consequences; verify anything you act on with a professional.

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# 1. How to read this

## The filter every source had to pass

A source qualifies only if at least **two** of these five hold:

1. runnable code exists;
2. it states transaction costs, slippage or fills explicitly;
3. the author has actually traded or built it, not merely written about it;
4. **it names its own failure modes and where the technique stops working** — this counts double;
5. it is reproducible on data you can actually obtain.

Auto-rejected regardless of anything else: any backtest presented with no cost model; anything whose primary funnel is a paid course, membership or signal service; "I made $X" and win-rate marketing; anonymous authors; and broker-affiliated "education" where order flow is the business model.

Test (4) carries double weight deliberately. A source that tells you where a technique breaks is worth three that tell you it works. It is also the test that most of the field fails.

## What the words mean

- **ALIVE** — still works in practice as of 2026, for someone in your position.
- **DECAYED** — the effect was real and has been arbitraged, crowded or regulated out.
- **Firm-only** — the technique works, but the barrier is capital, exchange membership or infrastructure you cannot assemble alone.
- **Unverified** — reported from search metadata, or from a source that could not be opened. A lead, never evidence.

## The one honest caveat about this document

Two rounds produced material of genuinely different quality. The strongest findings here come from agents that *measured something* — downloaded an NSE bhavcopy, queried a live exchange API, cloned a repository and read its cost code. The weakest are literature summaries. Where a finding rests on measurement, this document says what was measured and on what date, so you can re-run it. Where it rests on reading, it says so.

Prices, rates and market-structure facts were current on 27–28 August 2026, and several are on a known schedule to change. Section 3.5 lists the structural breaks explicitly, because at least three of them fall inside the data you are about to receive.

**The single most useful habit this research suggests**

Before believing any options result — yours or anyone's — find the line where the bid-ask half-spread is charged, and read the units. A published, peer-reviewed, code-shipping 0DTE paper had that line wrong by a factor of one hundred. When it was corrected, essentially every strategy in it went from positive to negative net Sharpe. Section 6.1 gives the anatomy.

# 2. The landscape map

Twenty-five fields, each with the same eight items, so you can compare them side by side and decide what to ignore. The eighth item is mandatory and is the reason this is a triage document rather than a menu: every survey fails by making everything sound worthwhile. A ranked shortlist closes the section.

## 2.1 Options and volatility

### 1. Arbitrage-free surface modelling (SVI / SSVI / local vol)

**Is** Parametric fitting of the implied vol surface subject to static no-arbitrage constraints, plus the Dupire local-vol bridge. Output is a clean, interpolable, differentiable surface.

**Uniquely solves** Every other options technique needs a surface that does not produce negative densities or fake calendar arbitrage at 09:20 when the chain is garbage. Nothing else gives you that.

**Status** **ALIVE as infrastructure**, **DECAYED as alpha**. Fitting SVI has not been a trade since roughly 2012. What survives is intraday refitting on minute data and quote-quality filtering.

**Prereqs** You have them. Infrastructure: a tick-to-surface pipeline and a fast constrained optimiser. Capital: zero.

**First useful result** 2–3 weeks, to convert your existing research artefact into a minute-wise NIFTY surface service with residual diagnostics.

**Solo?** Entirely.

**Sources** Gatheral & Jacquier, *Arbitrage-free SVI volatility surfaces* (arXiv:1204.0646); the Zeliade quasi-explicit calibration white paper, for the failure geometry rather than the formula; Bennett, *Trading Volatility*, free at `https://trading-volatility.com/Trading-Volatility.pdf`.

**Why this might waste your time**

You already did it, and a better surface fit has never by itself made anyone money — you are polishing a lens, not finding something to look at. Worse, Section 4.3 shows that on an unfiltered NSE chain your "residual" is substantially your model disagreeing with NSE's model.

### 2. Volatility risk premium harvesting (short vol)

**Is** Selling implied against subsequently realised volatility, systematically, hedged or unhedged. The most documented positive-expectancy structure in options.

**Uniquely solves** The only options return stream that is a compensated risk premium rather than a forecasting bet. It pays you for supplying insurance.

**Status** **DECAYED in the US**, **ALIVE and structurally distorted in India**. SPX VRP is the most crowded retail-adjacent trade alive and 0DTE has compressed it. Indian VRP is fatter, but the fat part is exactly the part that kills you.

**Prereqs** Minimal maths. What you actually need is a margin model and a survival-through-drawdown design. This is the one field where being small hurts.

**First useful result** 3–4 weeks to a cost-honest backtest. Years to know whether it survives.

**Solo?** Yes — and the risk of ruin is the highest of anything on this map.

**Sources** CBOE PUT / BXM / CNDR methodologies and full history, the public benchmark that makes an alpha claim falsifiable; SEBI's derivatives studies; Moontower (Kris Abdelmessih, 21 years market-making at SIG and Parallax).

**Why this might waste your time**

The edge is real, small and levered. Your backtest will show a beautiful Sharpe that is entirely a description of what has not happened yet. Per SEBI's August 2026 behaviour study, the option sellers who lose lose roughly ₹51.7 lakh each — about eleven times the buyers — on a 44% loss rate that looks reassuring right up until it is not.

### 3. Dispersion and correlation trading

**Is** Short index vol against long single-name vol, monetising the spread between index-implied and constituent-implied correlation.

**Uniquely solves** It isolates *correlation* as a tradeable asset. No directional or single-name view survives the construction.

**Status** **ALIVE but institutional** in the US; effectively **CLOSED in India**, and for structural reasons rather than liquidity ones. Section 7.6 gives the measured verdict.

**Prereqs** Correlation estimation, vega-weighting, a margin engine, and in India a physical-delivery operation across fifty names.

**First useful result** 6–10 weeks to a paper study that will not be tradeable.

**Solo?** No.

**Sources** Bennett's dispersion chapters, the best free treatment in existence; Bossu, *A Practical Guide to Dispersion*.

**Why this might waste your time**

You will build it, find that it works on paper, and then discover that Indian single-stock options are physically settled, that the expiry calendars no longer line up, and that SPAN gives you no correlation offset at all across underlyings — so you post margin as though you held fifty-one unrelated books.

### 4. Options market making

**Is** Continuous two-sided quoting with inventory-aware skew, where P&L is edge per trade minus adverse selection minus hedging cost.

**Uniquely solves** The only options activity where you are paid the spread rather than paying it. The sign of the cost term flips.

**Status** **ALIVE in crypto**, **DEAD to outsiders in Indian index options**, where quoting is a colocation race against sub-100-microsecond stacks.

**Prereqs** Avellaneda-Stoikov and successors, queue models, real exchange connectivity, inventory capital. Colocation for anything Indian.

**First useful result** 2–3 months on crypto at small size. Never, for NSE.

**Solo?** Crypto yes, at small size, on mid-cap perps. Indian index options no.

**Sources** `hummingbot`; `hftbacktest` (`https://github.com/nkaz001/hftbacktest`), which models queue position and feed *and* order latency separately; Avellaneda & Stoikov (2008) for the reservation-price logic, not the parameters.

**Why this might waste your time**

Market making is 10% model and 90% infrastructure, operations and exchange relationships. The part you enjoy is the part that matters least.

### 5. Options flow and dealer positioning (GEX, vanna, charm)

**Is** Inferring the dealer hedging book from open interest and trade-side classification, then predicting the mechanical hedging flow it forces.

**Uniquely solves** The only framework that predicts *flow* — a forced, non-discretionary buyer or seller — rather than value.

**Status** **Half-decayed in the US** and, as ordinarily constructed, **sign-inverted on NIFTY**. Round two refuted the standard construction against NSE's own published data; see Section 4.4. What remains is a harder and considerably more defensible project.

**Prereqs** Trade-side classification — the hard part — open-interest reconciliation, and a surface for the greeks. Capital: low.

**First useful result** 4–6 weeks, and only if you treat the dealer sign as a parameter to be estimated rather than an assumption to be imported.

**Solo?** Yes, and the Indian version is genuinely unpublished.

**Sources** NSE participant-wise open interest, daily, signed by category, at `nsearchives.nseindia.com/content/nsccl/` Barbon & Buraschi, *Gamma Fragility* — the free, named, testable version of the claim Karsan makes informally.

**Why this might waste your time**

The whole edifice rests on guessing which side of each trade the dealer took. Get that classification wrong and you have built a very sophisticated random number generator — and on NIFTY the standard guess is demonstrably backwards.

### 6. Event-driven volatility

**Is** Trading the term-structure kink around a scheduled event. Implied vol embeds an event premium that decays deterministically once the event passes.

**Uniquely solves** A dated, non-stationary edge with a known catalyst — the only options trade where you know in advance when you find out you were wrong.

**Status** **ALIVE**. The Indian equivalents — RBI policy, budget day, election day — are considerably less picked over than US mega-cap earnings.

**Prereqs** A clean event calendar, an event-vol extraction model, and brutal cost discipline. These are wide-spread trades.

**First useful result** 3–5 weeks.

**Solo?** Yes.

**Sources** Bennett's event chapters; Dubinsky & Johannes for the decomposition (academic, no cost model, so structure only); Klassen's *Event Modeling* talk for the two-jump Merton treatment.

**Why this might waste your time**

The premium is real, and the bid-ask is wider than the premium in exactly the names where the premium is largest.

### 7. Delta hedging, discrete-hedging error, and P&L attribution

**Is** The mechanics of turning an options position into a volatility position: hedge frequency, band choice, and the decomposition of realised P&L into theta, gamma, vega, vanna and residual.

**Uniquely solves** The only thing that tells you *why* you made money. Without attribution, every options result is unattributable luck.

**Status** **ALIVE, and permanently so.** Not an alpha field — it is the accounting layer that makes every other field legible, and the single most common thing solo traders skip.

**Prereqs** Minute data, which you are about to have; a greeks engine; a cost model.

**First useful result One to two weeks. The fastest payoff in this entire document.**

**Solo?** Trivially.

**Sources** Section 5 in full.

**Why this might waste your time**

It generates no signal whatsoever. It only tells you the truth about signals you already have, which is unglamorous and therefore usually skipped.

## 2.2 Microstructure, execution, speed

### 8. Market microstructure (order-book dynamics, adverse selection, queue)

**Is** How price forms from the order book: queue priority, order-flow imbalance, trade classification, and the information content of quotes.

**Uniquely solves** The only field that explains *why your fill differed from your backtest*. The gap between paper and live P&L lives entirely here.

**Status** **ALIVE and undervalued by solo traders**. Order-flow imbalance as a one-to-five-second predictor still works; it is not tradeable without low latency, but it is usable to improve entry timing on slower strategies.

**Prereqs** Level-2 or level-3 data. Your minute data is too coarse for this — it needs tick. Plus serious data engineering.

**First useful result** 6–8 weeks, gated entirely on data acquisition.

**Solo?** For research and execution improvement, yes. As standalone alpha, no.

**Sources** `hftbacktest`; Cont, Kukanov & Stoikov, *The Price Impact of Order Book Events*; Bouchaud et al., *Trades, Quotes and Prices* — and its free predecessor, Bouchaud, Farmer & Lillo, *How Markets Slowly Digest Changes in Supply and Demand* (arXiv:0809.0822), roughly 100 pages covering a large fraction of the book.

**Why this might waste your time**

All the real signal lives at horizons shorter than your latency. You will learn a great deal and be able to act on almost none of it.

### 9. Execution algorithms and transaction cost analysis

**Is** Scheduling a parent order into children to trade off market impact against price risk, and measuring the result against benchmarks.

**Uniquely solves** It converts a theoretical edge into a realised one. For a strategy with 5 bp of edge and 4 bp of cost, execution *is* the strategy.

**Status** **ALIVE, permanently.** Not crowded — it is a cost-reduction discipline, and cost reduction cannot be arbitraged away.

**Prereqs** Almgren-Chriss, an impact model calibrated to your own instruments, and honest fill logging.

**First useful result** 2–3 weeks for TCA on your own fills; months for a calibrated impact model.

**Solo?** Yes, and disproportionately valuable at small size in Indian options, where spreads dominate.

**Sources** Almgren & Chriss, *Optimal Execution of Portfolio Transactions*; Almgren et al., *Direct Estimation of Equity Market Impact*.

**Why this might waste your time**

At your size your orders have no market impact, so most of the literature's machinery is irrelevant. Only the spread-crossing decision matters.

### 10. High-frequency trading and latency arbitrage

**Is** Competing on speed for stale-quote and cross-venue discrepancies, measured in microseconds.

**Uniquely solves** Captures the mechanically riskless component of price dislocation. No forecast required.

**Status** **DEAD to you.** Not decayed — structurally inaccessible. It is a capital-expenditure race: colocation, FPGA, microwave. There is no evening-hours version.

**Prereqs** Seven figures of infrastructure and an exchange membership.

**First useful result** Never, as a solo.

**Solo?** Absolutely not.

**Sources** Budish, Cramton & Shim, *The High-Frequency Trading Arms Race*, which is the best argument for not entering; Aquilina, Budish & O'Neill for the FCA's message-level measurement.

**Why this might waste your time**

Reading about it is entertaining. There is a zero percent chance you can do it, and the reading crowds out fields where you can.

## 2.3 Systematic equity and cross-asset

### 11. Statistical arbitrage and mean reversion

**Is** Trading the residual of an asset against a hedged basket or partner, betting on reversion of a spread that has no reason to trend.

**Uniquely solves** Market-neutral return with a mechanically defined hedge — the only way to make the market factor genuinely disappear.

**Status** **Heavily decayed at daily frequency**, **alive intraday**. Classic daily pairs on liquid US equities has been flat for a decade.

**Prereqs** Cointegration, Ornstein-Uhlenbeck, Kalman filtering. In India, cash-market shorting is intraday-only and the stock-lending market is thin, so you cannot hold the short leg.

**First useful result** 3–4 weeks to a backtest, and the backtest will be wrong by roughly the size of the edge.

**Solo?** At small scale, partially.

**Sources** Avellaneda & Lee, *Statistical Arbitrage in the U.S. Equities Market*, which includes an explicit decay analysis; Robot Wealth, who publish negative results; Robert Carver, `https://qoppac.blogspot.com` and `pysystemtrade`, where costs are a first-class object in the codebase.

**Why this might waste your time**

Every published stat-arb result is a lower bound on how crowded it now is, and the surviving edge is smaller than the Indian STT-plus-impact cost stack.

### 12. Factor investing and cross-sectional equity

**Is** Sorting a cross-section on characteristics and holding the long-short spread.

**Uniquely solves** Long-horizon compounding with capacity — the only field here that scales to serious assets under management.

**Status** **DECAYED as alpha**, **alive as a benchmark**. The replication crisis gutted most of the zoo. Momentum and quality survive weakly.

**Prereqs** Panel-data hygiene and a survivorship-bias-free universe, which is the real barrier in India (see Section 7.6.1).

**First useful result** 4–6 weeks, mostly spent cleaning data.

**Solo?** Yes, but the payoff horizon is years and the Sharpe is around 0.4.

**Sources** Hou, Xue & Zhang, *Replicating Anomalies* — emphatically a list of things that do not work; Harvey, Liu & Zhu; AQR's published library.

**Why this might waste your time**

It is slow, low-Sharpe, capacity-oriented investing dressed as quant research — the opposite of a minute-wise options trader's comparative advantage.

### 13. Trend following and time-series momentum

**Is** Long-horizon directional betting on the sign of past returns across a diversified futures basket, sized by inverse volatility.

**Uniquely solves** Positive skew and crisis convexity. The only systematic strategy that reliably makes money when everything else is dying — directly relevant to a firm called Convex Hedge.

**Status** **ALIVE but degraded.** Sharpe has fallen from roughly 1.0 pre-2008 to 0.4–0.5. The convexity is the point, not the Sharpe.

**Prereqs** Low maths, high diversification requirement — 30-plus instruments, which means global futures. Capital is the binding constraint.

**First useful result** 2 weeks with `pysystemtrade`.

**Solo?** If you can access global futures. Borderline India-only.

**Sources** Carver (ex-AHL, runs it live, publishes his own failures); Moskowitz, Ooi & Pedersen; Hurst, Ooi & Pedersen for the century-long evidence. See also Section 9.5 for the trend-options identity, which is the part that connects to your actual work.

**Why this might waste your time**

It needs breadth and patience, both of which you currently lack. A trend book on three Indian instruments is not trend following; it is gambling with extra steps.

### 14. Portfolio construction and risk (covariance, sizing, Kelly)

**Is** Turning a set of signals into position sizes: covariance estimation, shrinkage, risk parity, hierarchical risk parity, leverage choice.

**Uniquely solves** It determines the *geometric* return, which no signal research touches. Two traders with identical signals and different sizing have unrelated outcomes.

**Status** **ALIVE, permanently, and systematically underrated.** Not crowded, because it is not a trade.

**Prereqs** Linear algebra, Ledoit-Wolf shrinkage, and an understanding of why mean-variance optimisers explode.

**First useful result** 2–3 weeks, and it immediately improves everything else you build.

**Solo?** Yes.

**Sources** Ledoit & Wolf; López de Prado on hierarchical risk parity, free on SSRN with code in the paper; and Section 5.3 for the options-specific treatment, which is the version you actually need.

**Why this might waste your time**

Optimal sizing of a nonexistent edge is exactly zero. It is very easy to spend three months here instead of finding the edge.

### 15. Research methodology and backtest hygiene

**Is** The statistics of strategy selection: multiple-testing correction, deflated Sharpe, walk-forward design, and cost modelling as a first-class citizen.

**Uniquely solves** The only defence against the dominant failure mode in solo quant work — believing a backtest.

**Status** **ALIVE, and the highest return per hour spent on this map.**

**Prereqs** Statistics you already have. Discipline you may not.

**First useful result** One weekend to implement probability of backtest overfitting and a trial counter.

**Solo?** Yes.

**Sources** Bailey & López de Prado on the deflated Sharpe ratio and on combinatorially symmetric cross-validation. Note the assessment in Section 6: implement **PBO via CSCV**, not DSR — DSR needs the number of trials, and nobody honestly reports their own.

**Why this might waste your time**

It only ever tells you that things you liked do not work. It produces no strategies, only funerals.

## 2.4 Machine learning and data

### 16. Machine learning applied to markets

**Is** Supervised and reinforcement models on financial features, with the special problems of non-stationarity, low signal-to-noise, overlapping labels and near-zero effective sample size.

**Uniquely solves** Nonlinear interaction discovery across many weak features, genuinely hard to do by hand once you have more than twenty.

**Status** **Mostly decayed at the solo level**, **alive narrowly**. It wins where the label is short-horizon and microstructural — next-100ms mid move, fill probability, queue depletion — and loses almost everywhere else. Deep learning on daily OHLCV is a well-documented dead end.

**Prereqs** You have the maths. The real prerequisite is purged and embargoed cross-validation, and enough independent samples — which minute options data gives you and daily data does not.

**First useful result** 6–10 weeks, most of it spent discovering leakage.

**Solo?** For narrow, well-posed problems, yes. End-to-end, no.

**Sources** López de Prado's free SSRN corpus — triple-barrier labelling, meta-labelling, purged CV, fractional differentiation. The book is paid; the constituent papers are free and carry nearly all the substance.

**Why this might waste your time**

The most seductive and least productive field here for a solo trader. You will build impressive pipelines whose out-of-sample Sharpe is indistinguishable from a moving-average crossover, and the sunk cost will make you keep going.

### 17. Alternative data

**Is** Non-price inputs — satellite, card spend, web scrape, filings text — mapped to fundamentals ahead of consensus.

**Uniquely solves** True informational rather than statistical edge: actually knowing something before the market does.

**Status** **Decayed at the top**, **alive in obscure corners**. Every mainstream dataset is licensed by two hundred funds and priced accordingly. What is alive for a solo is *self-collected* data nobody sells.

**Prereqs** Scraping, ETL, storage, legal care.

**First useful result** Three months or more, and mostly engineering.

**Solo?** Only for self-collected data.

**Sources** Round one could not verify a single source in this field that cleanly passes the filter and is free — most of the literature is vendor marketing. It is reported as **unsourced** rather than padded. The one exception worth naming: NSE's own free participant-wise and open-interest archives are alternative data that almost nobody parses properly.

**Why this might waste your time**

For a solo trader this is 90% data engineering and 10% finance, and the 10% is only valuable if you have a stock-picking process to feed — which you do not.

### 18. Research and data infrastructure

**Is** The unglamorous layer: tick capture, a time-series store, point-in-time correctness, corporate-action and contract-roll handling, deterministic backtest-live parity.

**Uniquely solves** The only thing that makes results reproducible and prevents look-ahead. Minute-wise Indian options data is large and messy; without this layer nothing else is trustworthy.

**Status** **ALIVE and mandatory.** Not a crowding question.

**Prereqs** Engineering. Parquet/Arrow, columnar storage, and a symbology scheme for expiring option contracts.

**First useful result** 3–4 weeks, and it gates every other field.

**Solo?** Yes. Section 6.5 gives a specific recommendation so you can stop thinking about it.

**Sources** NautilusTrader, for nanosecond event-driven design with identical strategy code in backtest and live; `hftbacktest`; ArcticDB.

**Why this might waste your time**

Infrastructure is infinitely extensible and has no natural stopping point. You can spend a year building a platform and zero days trading.

## 2.5 Crypto

### 19. Perpetual funding, basis, and cash-and-carry

**Is** Harvesting the funding rate on perpetual swaps by holding the offsetting spot or dated future, and trading the basis term structure.

**Uniquely solves** A mechanically defined, observable, high-frequency carry stream that exists nowhere else in finance. The funding rate is *printed* every eight hours.

**Status** **ALIVE but compressed.** BTC and ETH basis is heavily farmed by ETF-arbitrage desks; annualised carry has fallen from 20%-plus to single digits in calm regimes. What remains alive: altcoin funding, funding dislocations during liquidation cascades, and cross-exchange funding spreads.

**Prereqs** Low maths, high operational competence — exchange API risk, margin and liquidation modelling, custody, counterparty risk.

**First useful result** 2–3 weeks. The fastest path to a live, cost-honest, small-capital P&L on this map.

**Solo?** The most solo-viable strategy here — but see the tax verdict in Section 3.2, which reverses the conclusion for an Indian resident.

**Sources** Hummingbot's `v2_funding_rate_arb.py` — read issue #7295 first, which documents a double fill on Hyperliquid perps after the strategy is stopped; *Anatomy of the Oct 10–11 2025 Crypto Liquidation Cascade* (SSRN 5611392) as the stress scenario your backtest must survive.

**Why this might waste your time**

The carry is small and the tail is exchange insolvency, oracle failure, or a liquidation cascade that takes your hedge leg out. You are short a catastrophe option to earn about 6%. On 10–11 October 2025, auto-deleveraging forcibly closed market makers' short perp legs against their long spot — the exchange took away the profitable side of the delta-neutral book. That has no analogue on NSE, and any risk framework you carry over will not model it.

### 20. Crypto options and volatility

**Is** Volatility trading on Deribit and the Indian-registered venues, plus the DeFi analogue of option vaults and AMM liquidity provision as a short-gamma position.

**Uniquely solves** The place where your existing SVI/SSVI machinery transfers immediately to a market with fat spreads, no colocation requirement, 24/7 data, and few sophisticated participants relative to SPX.

**Status** **ALIVE — the least crowded serious options market a solo can reach**, with two large caveats measured in Section 8: the fee cap makes cheap options untradeable, and the Indian tax regime makes deployment uneconomic.

**Prereqs** Exactly what you already have, plus API plumbing and the inverse-contract correction.

**First useful result** 4–6 weeks. Your SVI work is most of the way there.

**Solo?** Yes. Strongest fit-to-skill ratio on the entire map — as *research*. Not as a P&L venue.

**Sources** Echenim, Gobet & Maurice (Kaiko), which proves that calibrating SVI on crypto mid-prices systematically underestimates wing vol and gives an anchor price to replace the mid; Magadini, Sinclair &

Chepal on real-time SVI calibration; Lucic & Sepp on inverse options with code at `https://github.com/ArturSepp/StochVolModels`.

**Why this might waste your time**

Deribit's liquidity is concentrated in a handful of BTC and ETH expiries. Outside them, the "opportunity" you find in the data is a quote nobody will trade against. And on sub-$200 options you pay 25% round-trip in fees plus a 24% spread, so your edge must exceed roughly half the premium.

### 21. On-chain analytics and MEV

**Is** Reading blockchain state directly — DEX pool composition, liquidations, mempool — and, for MEV, competing to reorder transactions.

**Uniquely solves** Complete, free, real-time visibility into a market's entire order flow and positioning, a condition that exists in no traditional market.

**Status** MEV is **decayed into a relay oligopoly**; on-chain *analytics* is **alive and cheap**. The measurement is decisive: over 2023–2025, three firms captured about 90% of extracted value by Q1 2025, and they transfer roughly 90% of arbitrage revenue to their integrated builders to buy auction priority — a subsidy you cannot match. One tracked searcher ran at $-272\%$ margin.

**Prereqs** EVM literacy and an indexer. MEV additionally needs colocated infrastructure and a relay relationship.

**First useful result** Analytics, 3–4 weeks. MEV, do not.

**Solo?** Analytics yes; MEV effectively no.

**Sources** *Measuring CEX-DEX Extracted Value and Searcher Profitability* (arXiv:2507.13023) — a measurement paper that will not teach you to build a searcher but will save you six months of trying.

**Why this might waste your time**

It is a different discipline wearing a finance costume. You would be learning distributed systems engineering, not trading. And the barrier is not skill, it is a commercial builder relationship.

### 22. Volatility harvesting and the rebalancing premium

**Is** Rebalancing between a volatile asset and cash to capture the difference between arithmetic and geometric mean return.

**Uniquely solves** Nothing, for you.

**Status** **Retail folklore, and the most misunderstood item here.** Shannon's Demon is real arithmetic, but the "rebalancing bonus" is payment for *selling variance and providing liquidity* — the payoff is short-gamma and mean-reversion-dependent, and it inverts in a trending market.

**Prereqs** None, which is the problem.

**First useful result** Immediately, and it will be wrong.

**Solo?** Yes, and that is the problem.

**Sources** Witte, *Volatility Harvesting* (arXiv:1508.05241) — read it to get the maths right, then stop.

**Why this might waste your time**

In crypto it fails twice over: taker fees on every rebalance, and BTC's realised drift has historically dominated the harvest term, making it an expensive way to underperform holding. Note also the structural conflict with trend following — they are the same bet in opposite directions, and running both without an explicit regime allocator means paying two spreads to hold no position.

## 2.6 India-specific

### 23. Indian market structure and the regulatory regime

**Is** The rules that determine whether an Indian strategy is viable at all: STT, stamp duty, position limits, SPAN and ELM margining, lot sizes, and the expiry calendar.

**Uniquely solves** The only thing that converts a US-derived technique into something that survives contact with NSE. Costs in India are not a rounding adjustment; they are a structural asymmetry that changes which structures are viable.

**Status** **ALIVE and actively changing.** This is time-sensitive knowledge. Section 3.5 lists five structural breaks in the last twenty-four months, at least three of which fall inside your data.

**Prereqs** None mathematical. Pure diligence.

**First useful result** One week to build a correct cost-and-calendar model. It will invalidate several strategies immediately, which is the point.

**Solo?** Mandatory, actually.

**Sources** SEBI circulars and NSE circulars, all free and machine-indexed — see Appendix 14.2. **Explicitly rejected:** Zerodha Varsity, Sensibull education, and every broker "learn" portal. They are often factually accurate on mechanics; use them as a mechanics reference, never as a source of a conclusion.

**Why this might waste your time**

It is compliance reading, not research, and it goes stale — SEBI changes the rules faster than you can encode them. The mitigation is architectural: put every rate in a dated configuration file, never in source. See the audit in Section 7.5 for what happens when you do not.

### 24. Indian index-options intraday structure

**Is** The specific behaviour of NIFTY and Sensex options intraday: the open volatility spike, midday decay, expiry-day gamma compression, and the retail flow signature.

**Uniquely solves** The only field where your data advantage is a genuine edge over the global research population, because almost nobody outside India studies this and almost nobody inside India studies it rigorously.

**Status** **ALIVE and genuinely under-researched.** Retail participation is enormous, documented, and loses money — so there is identifiable flow to be on the other side of. The crowded part is the naive expiry-day short straddle, which every algo desk in Mumbai is farming.

**Prereqs** Your existing stack plus the cost and calendar model from field 23.

**First useful result** 3–4 weeks. The fastest path to something original.

**Solo?** Yes, and this is where solo is *advantaged*.

**Sources** SEBI's FY25–FY26 studies for the shape of the losing flow; NSE bhavcopy and participant-wise archives, free and primary. Round two searched hard and could not find a single free, code-backed, cost-explicit research source on Indian intraday options microstructure. That vacuum is the opportunity, but it means nobody will hand you a template.

**Why this might waste your time**

"Under-researched" and "unprofitable" look identical from the outside, and there is a real chance the reason nobody published is that the costs eat everything. Section 7.1 gives you the number to check that against before you start.

## 2.7 Adjacent

### 25. Tail hedging and convexity

**Is** Deliberately paying negative carry for a payoff that is convex in a crash — deep OTM puts, VIX call spreads, long-vol overlays — sized so the cost is survivable.

**Uniquely solves** The only construction that pays off when correlations go to one and every diversification assumption breaks at once.

**Status** **ALIVE as a portfolio function**, **decayed as a standalone**. The famous 2020 results are survivorship-selected; most tail funds bleed. It works only as an overlay whose size is chosen by portfolio maths, not conviction.

**Prereqs** Convexity decomposition and honest accounting of the bleed.

**First useful result** 3–4 weeks — and it is directly on-mission for Convex Hedge.

**Solo?** As research yes; it needs a portfolio to overlay onto.

**Sources** Bennett on the cheapest way to buy a crash; Bhansali's free SSRN papers on tail-hedging cost, candid about the drag; trend following as a cheaper convexity source.

**Why this might waste your time**

Tail hedges lose money 95% of the time by design, and there is no backtest length that will make you confident you sized it right.

### Also considered, and set aside

**Fixed income and rates volatility** — the swaption vol cube is the only vol market with a genuine third dimension and the modelling problem is strictly richer than equity vol. It is also OTC, dealer-intermediated, with no retail access and no obtainable data, so it fails test (5) outright for you. Hagan et al., *Managing Smile Risk*, remains worth reading — but for its diagnosis of why model deltas are unstable, which transfers, rather than for SABR, which is inferior to what you already have. *Waste warning: you cannot get the data or the access, so it is pure intellectual tourism.*

## 2.8 The shortlist

Ranked for your position specifically — someone with the surface maths already done, minute-wise NIFTY data arriving, a full-time job, and no realistic access to US venues.

1. **Delta hedging, hedging bands, and P&L attribution** (field 7). Cheapest, fastest, and it is the measurement instrument for everything else. One to two weeks. Without it every subsequent result you produce is unattributable. Do this first, and in parallel with nothing.
2. **Realised volatility from minute data, and the deseasonalised-RV versus remaining-implied-variance study** (fields 24 and 2). This is the project where your data advantage is actually spent, it is invisible in end-of-day data, and Section 5.2.5 specifies it step by step. It is also genuinely publishable.
3. **Indian cost, calendar and margin modelling** (field 23). Not a strategy — a filter. It will kill several ideas you currently believe in, which is worth more than any new idea. One week, and Section 7.4 has most of it already computed.
4. **The SVI residual falsification test** (fields 1 and 24). Converts your existing codebase from infrastructure into a falsifiable hypothesis. The pre-registered specification is in Section 4.3; the honest expectation is that it fails, and finding that out in weeks rather than months is the whole value.
5. **Dealer positioning on NIFTY, rebuilt properly** (field 5). Higher risk than the others and a harder project than round one implied, but it is the one intersection where your data, a free daily primary file with no US equivalent, and a genuine research vacuum all coincide.

**Explicitly deprioritised despite being interesting:** machine learning applied to markets (highest seduction-to-yield ratio here), HFT and MEV (structurally inaccessible), dispersion (structurally closed in India), factor investing (wrong horizon for your data), rates vol (no data access), volatility harvesting (folklore).

# 3. Four findings that reshape the plan

These came out of round two's verification pass. Each was asserted or assumed in round one; each was checked against a primary document or against live data; and each changes what is worth building rather than merely adding detail.

## 3.1 The US is a source of technique, not a venue

**Verdict: REFUTED — and the received wisdom is wrong in a way that matters**

An Indian resident individual has **no LRS route to a US-listed option**, long or short. The chain closes without a gap, and it does not run through the sentence everyone quotes.

The derivation, from primary text:

1. **RBI Master Direction on LRS** (FED Master Direction 7/2015-16), para 6(iii): the exhaustive list of permissible *capital account* transactions runs only through Overseas Direct Investment and Overseas Portfolio Investment.
2. **FEM (Overseas Investment) Directions, 2022**, definition (ix)(a): "OPI shall not be made in … **any derivatives unless otherwise permitted by Reserve Bank**."
3. A US-listed option is a derivative. Therefore no LRS capital-account route exists. Separately, para 5 independently bars "remittance for margins or margin calls to overseas exchanges", which kills short options by a second route.

**The correction to the received wisdom.** Many advisory sites quote the RBI LRS FAQ as saying "remittance under the Scheme is not available for margin trading or derivative transactions." Round two downloaded the full FAQ and grepped it: the strings "derivative" and "option" appear **zero times**. The conclusion is right; the citation everybody uses is fabricated. If you ever argue this with a bank's authorised-dealer desk, cite definition (ix)(a)(iii) of the OI Directions — quoting a sentence that does not exist will lose the argument.

**GIFT City does not cure it.** The July 2024 circular widened IFSC remittance to "all permissible purposes under LRS" — which imports the same OPI carve-out. IFSC relocates the account; it does not create a derivatives permission. NSE International Exchange's own Global Access platform says so directly, excluding derivatives and crypto "due to regulatory restrictions under LRS norms."

**What this changes.** Everything you build must clear on NSE or on a domestically registered crypto venue. This belongs at the top of your plan, not in a footnote. The extensive US options literature in this document is technique transfer only, and you should read it that way — including the parameters, which will not transfer even where the methods do.

Could not be verified: no RBI document says in terms "a resident individual may not trade overseas exchange-traded options." No such sentence appears to exist, which is precisely why secondary sources invent one. Separately unresolved and worth a professional opinion: funds already held abroad and never remitted under LRS sit outside the remittance question but inside FEMA's residency net.

## 3.2 Crypto tax applies transaction by transaction, and that is a strategy filter

**Verdict: CONFIRMED transaction-level. No netting. And the section number in circulation is obsolete.**

Section 115BBH no longer exists. The Income-tax Act, 2025 replaced the 1961 Act with effect from 1 April 2026; the successor is **s.194(1), Table serial number 4**, with withholding at **s.393(1), Table Sl. 8(vi)**.

The operative text carries two independent prohibitions. Condition (a) forbids set-off "in computing the income referred to in column C" — that is what kills same-asset netting. Condition (b) then forbids set-off against anything else and bars carry-forward. Note that (b) says "any provision", not "any *other* provision"; the narrowing word is absent, which closes the inter-asset route.

Three converging confirmations were found, and the third is the strongest because it is not an interpretation at all:

- The statutory text itself.
- A ministerial answer in the Lok Sabha (21 March 2022) stating directly that loss from the transfer of one VDA will not be allowed to be set off against income from transfer of another. Flagged: sansad.in returned HTTP 500 on every URL pattern; this quotation is from search-result text of that answer, not from the PDF.
- **The filing utility enforces it.** CBDT's ITR-2 Validation Rules for AY 2025-26, Rules 201–202, require the Schedule VDA total field — literally labelled "sum of all *positive* incomes" — to equal the plain sum of column 7. That can only hold if every row is non-negative. **The software structurally cannot accept a negative row.** Reporting is transaction-wise.

### What that does to the arithmetic

Round two derived the general result rather than asserting it. Let a round trip have P&L $X$ with mean $\mu$, standard deviation $\sigma$, and define the per-trade information ratio $r = \mu/\sigma$. Tax is levied on the positive part only, with no cost deduction:

$$\text{after-tax per trade} \;=\; \mu - 0.312\big(\mu\,\Phi(r) + \sigma\,\varphi(r)\big)$$

Setting this to zero gives $r^* = 0.1491$. A trade must carry a per-trade information ratio of about **0.149** before the regime lets you keep anything. Equivalently, the annualised Sharpe needed merely to break even is $0.149\sqrt{N}$:

| Trades per year | Annual Sharpe needed to break even |
|---:|---:|
| 250 | 2.4 |
| 1,000 | 4.7 |
| 5,000 | 10.5 |
| 20,000 | 21.1 |
| 50,000 | 33.3 |

The corresponding number under the F&O regime is **zero**. Expressed as effective tax on gross edge, the regime charges **79%** at $r = 0.2$, **141%** at $r = 0.1$, and **265%** at $r = 0.05$. Only above roughly $r \approx 1$ does the 31.2% headline rate approximate reality.

**Every delta-neutral options strategy lives between $r = 0.02$ and $r = 0.3$.** In that band this is not a 30% tax; it is a 79%-to-640% tax.

The 1% withholding compounds it on a separate axis. With gross edge $e$ as a fraction of transfer consideration, withholding as a share of gross edge is $0.01/e$ — so the book breaks the moment gross edge falls below **1% of turnover**. Market making, basis capture, funding harvesting and gamma scalping earn 1–30 bp of turnover, so they break by a factor of three to a hundred on the withholding alone, before any tax is assessed.

**The conclusion, stated as arithmetic rather than opinion**

The F&O regime is *tax-transparent to strategy design*: break-even gross edge equals the cost stack, and tax is a scalar applied afterwards. The VDA regime is *not* — because the base is $\mathbb{E}[\max(X, 0)]$ rather than $\mathbb{E}[X]$, it changes the *ranking* of strategies, and it is close to maximally hostile to exactly the delta-neutral options work you are training to do. Build the research stack, execution stack, margin simulator and cost model against NSE. Treat crypto as a source of data and technique, and as a place to validate models — not as a P&L venue.

Genuinely unresolved: whether the 1% withholding attaches to VDA *derivatives*. A cash-settled perpetual or option on a VDA is not, on the face of the words, a transfer *of* a VDA, and no CBDT circular resolving it was found. The no-loss-offset half applies regardless. This is a gap in the law, not a research failure.

## 3.3 There is no NIFTY term structure to trade

SEBI's August 2026 study — published one week before this document, and superseding the FY25 figures round one used — reports the turnover concentration by time to expiry:

| Share of index-option turnover | FY25 | FY26 |
|---|---|---|
| On the expiry day itself (0DTE) | 70% | 59% |
| Within one day of expiry | 80% | 75% |
| Within seven days of expiry | 98% | 97% |
| Beyond ten days | about 1% | |

**Your surface has, in practice, one to two usable maturity points.** Calendars, diagonals, term-structure relative value and any SSVI calendar-arbitrage work have almost nothing to trade on this venue. This kills a whole class of ideas cheaply, which is the most valuable thing a survey can do, and it should reshape what you build before you write a line of code.

It compounds with a second structural fact: BANKNIFTY, FINNIFTY and MIDCPNIFTY weeklies were withdrawn in November 2024, so BANKNIFTY is monthly-only and has roughly three usable maturities. Any term-structure research on it is structurally unavailable, not merely thin.

## 3.4 Half the NSE option chain is model output, not market data

This is the finding that decides whether your SVI work contains a trade, and round two settled it by downloading the NSE F&O bhavcopy for 27 August 2026 (NIFTY spot 24,090.85; 1,590 NIFTY option lines across 18 expiries) rather than arguing about it.

| | $n$ | Settlement on the ₹0.05 tick grid | Close $\neq$ settlement |
|---|---|---|---|
| Traded (volume $>0$) | 769 | 90.5% | 12% |
| Untraded (volume $=0$) | 821 | **20.6%** | **100%** |

**821 of 1,590 lines — 51.6% — did not trade at all**, and **79.4% of untraded settlement prices sit off the five-paise tick grid entirely**, so they cannot be observed prices. NSE's contract specifications confirm the mechanism: an option not traded during the day is closed at its **Black-Scholes theoretical price**.

The imputation signature is visible directly. On the 29 September 2026 expiry, four consecutive zero-volume call strikes returned implied vols of **17.30, 17.21, 17.13, 17.00** — a near-constant, monotone, near-flat sequence sitting five to six vol points above their traded neighbours at 11.22% and 14.11%. That is not a smile. That is a flat-volatility model writing its own answer into the data.

**Why this might waste your time**

Fit SVI to an unfiltered NSE chain and you will fit NSE's constant-volatility assumption in the wings, then discover a "residual" that is your model disagreeing with theirs. Any surface work that uses end-of-day closes without filtering on traded volume is partly fitting NSE's own Black-Scholes output back to itself — which is why your wing fits will look far better than reality warrants.

**And the tick-quantization explanation is wrong.** Round one blamed tick size. Round two computed the implied-vol uncertainty from half a tick ($\Delta\text{IV} = 0.025/\text{vega}$) across the whole liquid wing of the five-day expiry: it runs **0.00 to 0.17 vol points**, against an IV range of 6.4% to 33% across that same wing. A half-tick is under 0.5% of the structure being measured. Zero contracts settled at or below ₹0.05 on the front expiry, so the "zero bid" concern does not bind at end of day either. Right conclusion, wrong mechanism — and the distinction matters, because it tells you the fix is a *volume filter*, not a wider error bar.

## 3.5 The structural breaks in your data

At least five, and three of them fall inside the period you are about to receive. Any backtest spanning one of these without explicit regime-splitting is pooling two different markets.

| Date | Change | What breaks if you ignore it |
|---|---|---|
| 20 Nov 2024 | Contract size raised to ₹15–20 lakh; weeklies cut to one benchmark index per exchange; +2% ELM on short options on expiry day | BANKNIFTY / FINNIFTY / MIDCPNIFTY weeklies *terminate* on 13, 19 and 18 Nov 2024. The series does not thin out — it ends. Margin on expiry day rises by rule. |
| 1 Feb 2025 | Upfront premium collection; calendar-spread margin benefit removed on expiry day | A position that is margin-efficient for six days becomes margin-expensive on the seventh, predictably and by rule. |
| 1 Sep 2025 | NSE expiry moves Thursday → Tuesday; BSE moves Tuesday → Thursday | Day-of-week seasonality, DTE mapping and expiry-day gamma all shift. Cross-exchange work spanning this date crosses a *swap*, not a one-sided move. **And the expiry dates of already-live contracts were rewritten in the contract master on 28 Aug 2025** — so if you rebuild your universe from today's masters, contracts that traded for weeks under Thursday expiry will present as Tuesday. Snapshot the contract master per date; never reconstruct it. |
| 28 Oct 2025 | Lot sizes revised: NIFTY 75→65, BANKNIFTY 35→30, FINNIFTY 65→60, MIDCPNIFTY 140→120 | Every notional, every margin, every per-lot cost. NIFTY's full history is 25 → 75 (Nov 2024) → 65. A P&L-per-lot series crossing these is wrong by 3× and then 13%. |
| 1 Apr 2026 | STT raised again: options 0.10%→**0.15%**, futures 0.02%→**0.05%**, exercise 0.125%→**0.15%** | The options round-trip cost floor rose 27%; futures rose 46%. Any cost model carrying the Oct-2024 rates is one revision stale. |
| 3 Aug 2026 | **Closing Auction Session** live, applied first to exactly the stocks that have derivatives | The NIFTY close — and therefore the final settlement value of every index option — is now derived from single-print auction equilibrium prices of the constituents, not from any VWAP. CAS runs 15:15–15:35; equity derivatives trade to 15:40. Expiry-day strategies calibrated on pre-August-2026 data are calibrated on a mechanism that no longer exists. |

**The settlement rule, resolved — and it is neither answer round one had**

NSE Clearing states it plainly and the two rules differ: **index** derivatives settle on the **close price of the underlying index**; **single-stock** derivatives settle on the **last-30-minute VWAP of the security**. The widely repeated claim that NIFTY settles on a 15:00–15:30 VWAP of the index is **wrong**. The confusion arises because the index close was historically computed from constituent closes, each of which *was* a last-30-minute VWAP — but a weighted sum of per-stock VWAPs is not the VWAP of the index series. If you have been reconstructing settlement by VWAP-ing the index tick series, you have computed the wrong number for every expiry. And since 3 August 2026 even that is superseded by the auction.

# 4. The volatility surface, pointed forward

You can already fit SVI and SSVI. This section is about the two things you cannot do yet: turn a fitted surface into a position, and know whether the fit means anything on the chain you actually have.

## 4.1 From a fitted surface to an actual trade

### The definition of "rich" that survives contact with a market

The single most useful reframing came from Kris Abdelmessih (SIG, then a relative-value commodities vol book at Parallax): fair value is *where you can actually transact*, not where your model says. A one-vol residual inside a three-vol-wide market is not an edge; it is a quote.

That converts the residual question from "how big is the mispricing" into "how big is the mispricing relative to the width of the market at that strike", which is a different and much harsher test.

### The error bar is the whole game

Timothy Klassen's Vola Dynamics material is the best free source on this, and its central design decision is the one to copy: **attach an error bar to every fitted vol**, widening automatically as quote quality degrades. A tradeable residual is one that exceeds the fitter's own uncertainty at that strike — not one that exceeds a fixed two-vol-point rule.

Round two added a second term that round one omitted entirely, and it matters:

**The error bar has two components, not one**

An SVI slice fitted to $n$ quotes has *parameter* uncertainty as well as *quote* uncertainty. The residual at strike $k$ must be compared against the combined quote uncertainty and fit uncertainty at $k$. Comparing against quote uncertainty alone is exactly how a fitted-surface artefact gets mistaken for signal.

Klassen is also explicit that **the mid is the wrong input in the wings**. His KOSPI example shows puts quoting 0.02 by 0.03 next to 0.01 by 0.02 — tick quantization that makes wing mid-IVs pure artefact. Zero-bid options drag mid-IV down in both wings; a naive fitter chases it, a robust one does not. Low-latency desks feed size-weighted or book-depth microprices with temporal smoothing instead.

And the observation that should make you cautious about your own data source: **settlement prices for illiquid series are themselves produced by a fitted surface**, so a residual measured against exchange settlement data is partly measuring someone else's fitter. Section 3.4 shows this is not hypothetical on NSE — it is 51.6% of the chain.

### What the structures actually express

Bennett's *Trading Volatility* (free PDF; the author ran derivatives strategy at Santander and Barclays) is the structural half: when a calendar expresses a term-structure view and when it is merely a vega/theta mismatch; why risk reversals and put-spread financing behave differently as skew moves; how a skew position bleeds through spot moves rather than through skew itself.

Its limits are worth stating: European institutional index vol, roughly 2012 vintage, no code, no cost model worth the name, and nothing on weeklies or 0DTE. It tells you what trade to put on, never whether the edge survives your spread.

**Why this might waste your time**

The academic option-return-predictability literature looks like the missing link here and is not. The best of it (Bali et al., RFS 2023, 12 million delta-hedged observations, 265 characteristics) reports results gross *and* net of costs — but the effective-versus-quoted-spread assumption does most of the work in the "profits survive costs" claim, and at full quoted spread much of it does not. Treat it as a cost-sensitivity template, not a strategy. It is monthly rebalancing on US single-name options, which you cannot trade anyway.

## 4.2 Does SVI even work on NIFTY weeklies?

**The finding first: nobody has published this, and that is real open ground**

An extensive search for practitioner work, Indian quant blogs and academic literature applying SVI or SSVI to NIFTY, BankNifty or Sensex turned up: Indian academic IV literature that is descriptive rather than parametric; one hackathon repository; one first-hand LinkedIn field report; and Vola Dynamics inviting you to "contact us" about fitting NIFTY, i.e. the people who actually solve this treat it as a commercial product and publish nothing.

The one genuine field report is worth quoting for its texture: a developer who built a live NIFTY SVI surface reports that NSE chain data is "far messier than I initially assumed", strikes sparse or stale, raw fits jagged and unstable near spot, so he warm-starts each slice from the previous solution to stop tick-to-tick jitter and weights ATM strikes more heavily. Verifiable person, no institutional backing, no cost model. A field report, not a result — but it matches everything else here.

### Four structural facts that decide the answer

1. **The forward breaks before SVI ever runs.** NIFTY weekly options have no matching weekly future. Take the forward from the nearest monthly future — or worse from spot with a dividend-yield guess — and the error enters as an apparent put-call IV asymmetry that looks exactly like a skew dislocation. **Imply the forward per expiry from put-call parity on the liquid near-ATM strikes, and fit OTM-only.** Otherwise your best-looking residuals are your own forward error.
2. **Wing quotes are tick-quantized to ₹0.05.** A NIFTY weekly put quoting 0.05 by 0.10 is a 67% relative spread; in vol terms the bid-ask straddles tens of vol points. The wings of a NIFTY weekly do not contain a fittable vol level at all, and fitting SVI's linear-in-$|k|$ wings to them is fitting the tick grid. *Note the nuance from Section 3.4: at end of day, in the liquid band, the half-tick contributes under 0.17 vol points. The tick problem is real in the far wings and on live quotes; it is not what contaminates the liquid range. The dominant contaminant there is imputation.*
3. **The front slice is 0–2 DTE most of the time.** Total variance $w = \sigma^2 T$ is of order $10^{-4}$ there. The SVI loss surface becomes nearly flat in $\rho$ versus $m$, and calibration pins $\rho$ at its bound — an identifiability failure that appears even on clean synthetic chains. SVI's asymptotically linear wings are a large-$T$ argument; at one day to expiry the true smile is jump-dominated and V-shaped near the money.
4. **There is no term structure** (Section 3.3). BANKNIFTY lost its weeklies in November 2024 and has roughly three usable maturities; NIFTY has 97% of turnover inside seven days. SSVI calendar work has nothing to calibrate against.

### What to do instead

Stated as a recommendation, not a finding: fit SVI **only on the band where the market genuinely quotes two-sided size** — roughly 10-delta to 10-delta on NIFTY weeklies, tighter on expiry day — in total variance, on OTM-only quotes, with a width-based weight, and **extrapolate the wings rather than fit them**. Publish an error bar per strike from the quote width, and refuse to call anything rich or cheap unless the residual exceeds it.

Anyone claiming a wing dislocation on a NIFTY weekly put quoting 0.05 by 0.10 is reporting a tick, not a trade.

## 4.3 The residual: is your codebase a trade or infrastructure?

Round one's two agents contradicted each other directly on this, and it is the question that decides what your existing work is worth. One said the surviving use of SVI is that "the fit residual itself is the signal." The other said the wings are measurement artefact. Round two adjudicated against real data.

**The adjudication**

**Both were partly right, and both stated the wrong mechanism.** The artefact is confirmed — but it comes from *imputed settlement prices* (79.4% of untraded settles are off the tick grid), not from tick quantization (under 0.17 vol points, less than 0.5% of the signal). And the residual claim is refuted as stated, for a different reason: a residual measured against a surface fitted to a chain that is half imputed is not a market signal. At end of day the signal and the contaminant are not separable.
**Your SVI codebase is infrastructure, not a trade — as currently fed.** It becomes a candidate trade only under filters neither round-one agent specified.

### The quote-quality filters, all mandatory, all falsifiable

1. **Drop every strike with zero traded volume in the bar.** This alone removes about 52% of the end-of-day chain and all of the imputation signature.
2. **Never fit to settlement or close price.** Both are contaminated for untraded strikes and close is additionally stale. Fit to intraday two-sided quotes — which is exactly what your incoming minute data provides and what the end-of-day bhavcopy structurally cannot.
3. **Require a two-sided quote with bid $> 0$** and a relative spread below a threshold set from the data, not from taste.
4. **Screen on the tick grid as a permanent integrity check.** Any price off the five-paise grid in an intraday feed is imputed or erroneous. Free, decisive, and it should run always.
5. **Restrict to $|\log$-moneyness$| \leq 0.10$ on the front weekly**, widened only where measured volume supports it.
6. **Front one or two expiries only.** Per SEBI, 97% of turnover is within seven days and 1% beyond ten. Any calendar or term-structure residual strategy is untradeable on this venue — a hard finding that kills a class of ideas cheaply.

### The pre-registered falsification test

Write this down before you look at the data.

**The test**

The residual is signal *if and only if* the sign of today's residual at strike $k$ predicts the sign of tomorrow's change in that strike's mid-IV, out of sample, **after subtracting the round-trip half-spread at that strike**, on strikes that passed filters 1–6.

Three specific ways it dies. Check each; do not wait to discover them.

- **It reverses.** The residual predicts mean reversion *of the fit*, not of the market — it forecasts its own next recalibration. Test by refitting with a different SVI parameterisation (raw versus SVI-JW) and checking the residual survives. If it is parameterisation-dependent, it is an artefact of your own model.
- **It is real but uncapturable.** The residual is smaller than the half-spread at every strike where it is statistically distinguishable from zero. **This is the modal outcome**, and it is precisely what happened to the Vilkov 0DTE results (Section 6.1) — an entire result set inverted on a cost term two orders of magnitude smaller than a NIFTY wing spread. Design to detect this early.
- **It is a liquidity proxy.** The residual is largest exactly where volume is lowest, meaning you have rediscovered the illiquidity premium. Test by regressing residual magnitude on contemporaneous volume and spread; if those two absorb it, there is nothing left.

**What changes on the answer.** If it survives all three, your SVI work is the core of a trade and everything else is plumbing. If it does not, the codebase is still worth having — as the surface that prices and risk-manages positions you take for other reasons — but it is not the alpha, and you should stop treating it as such. Either way the question is **intraday-only**: it cannot be resolved on end-of-day data, and your incoming minute-wise NIFTY feed is the correct and possibly the only instrument. That is the good news. The decisive experiment is one you can actually run, in weeks.

## 4.4 Dealer positioning: refuted as stated, and the better door it opens

Round one asserted that dealer-positioning proxies are computable from open interest plus a Black-Scholes gamma map, and presented this as the one implementable framework in that report. Round two made the identifying assumption explicit and then tested it.

### The assumption, stated

The standard GEX construction computes $\text{Spot}^2 \times \Gamma \times \text{OI}$ per strike, with **call OI positive and put OI negative**. That sign convention is not a fact about the market. It is an assumption that *dealers are long every call and short every put* — equivalently, that customers buy puts for protection and sell calls in overwriting programmes.

Open interest is *unsigned*. Every contract has a long and a short, and the exchange publishes only the total. Nothing in an OI file distinguishes a dealer's long from a customer's long. In US index options the assumption is a defensible stylised fact grounded in the protection-buying bid. It is empirical, not structural, and it carries all the weight.

Studies that actually identify dealer sign — Ni, Pearson, Poteshman and White; Baltussen et al. — use **signed order-flow data** (CBOE/ISE Open-Close, which flags buy-to-open versus sell-to-open), not open interest. That is the honest state of the art: sign comes from flow data.

### On NIFTY the assumption is contradicted, and the primary file is free

NSE publishes signed participant-category option positions *daily*, at `nsearchives.nseindia.com/content/nsccl/f` with columns for Option Index Call Long, Put Long, Call Short and Put Short by client type. **There is no US equivalent at this granularity, freely published, daily.** Neither round-one agent mentioned it.

Three consecutive days, net positions in contracts:

| Date | Participant | Net calls | Net puts |
|---|---|---|---|
| 25 Aug 2026 | Client | $+94{,}693$ | $\mathbf{-604{,}022}$ |
| | FII | $-198{,}430$ | $+549{,}436$ |
| | Pro | $+99{,}801$ | $+24{,}124$ |
| 26 Aug 2026 | Client | $+191{,}071$ | $\mathbf{-663{,}004}$ |
| | FII | $-235{,}885$ | $+555{,}751$ |
| 27 Aug 2026 | Client | $+289{,}153$ | $\mathbf{-760{,}983}$ |
| | FII | $-265{,}856$ | $+626{,}360$ |

**Read the put column.** The standard convention requires customers to be net *long* puts. On NIFTY, clients are net **short** puts by 600,000 to 760,000 contracts, consistently and growing, while FIIs are net long puts and net short calls. The retail put leg has *the opposite sign* to the US stylised fact. Applying the standard GEX sign convention to NIFTY does not merely lack evidence — it inverts the sign of the dominant leg.

### What the file does and does not give you

It is aggregated across all strikes and expiries, across all index options rather than NIFTY alone, daily rather than intraday, and "Pro" is proprietary trading members — a proxy for market makers, not identical to them. So it does not hand you dealer gamma.

What it does is **falsify the sign assumption and give you a daily aggregate constraint**. Fit whatever strike-level sign model you like, then check that its aggregate reproduces this file. That test is free, daily, and nobody in either round proposed it.

Has anyone shown out-of-sample value for GEX on NIFTY? No — a genuine "nobody has published on this". The search returned only commercial dashboards, all of which auto-reject on the paid-funnel and no-cost-model criteria, and none of which disclose their sign assumption. The academic literature is entirely S&P 500, and the honest null there is a pre-registered SPY backtest over 1,972 days finding that GEX and its relatives add little after controlling for VIX and ATM implied vol. Benchmark anything you build against those two controls before believing it.

# 5. The measurement layer

Three techniques, all closed-form or near it, all implementable in weeks, all absent from round one. This is the fastest-payoff section in the document and the one that makes every other result you produce legible.

## 5.1 How often should you hedge?

### The literature, in one notation

Zakamouline's free 2005 working paper is the single most useful document here because it contains all five strategies in one consistent notation *plus* the horse-race between them:

- **Leland**: inflate volatility, $\sigma_m^2 = \sigma^2(1 + \sqrt{8/\pi}\,\lambda/(\sigma\sqrt{\Delta t}))$, then hedge at the resulting delta. Note its own failure: as $\Delta t \to 0$ the modified volatility is unbounded and the hedge degenerates to holding one share.
- **Whalley–Wilmott**: no-trade band of half-width $H = \left(\frac{3}{2}e^{-r(T-t)}\lambda S\Gamma^2/\gamma\right)^{1/3}$ around the Black-Scholes delta. **The cube root is the signature of the asymptotic solution.** No volatility adjustment.
- **Barles–Soner**: band plus a gamma-dependent vol adjustment.
- **Hodges–Neuberger** and **Davis–Panas–Zariphopoulou**: the exponential-utility formulation, solved numerically. Davis et al.'s useful result is that the option price is approximately invariant to the *form* of the utility function — mainly only the level of risk aversion matters.
- **Zakamouline's approximation**: adds a term $H_0 = \lambda/(\gamma S\sigma^2(T-t))$ that the two asymptotic strategies lack entirely.

His Monte Carlo (100,000 paths, 250 rebalance opportunities) ranks them: approximation > Whalley–Wilmott > Barles–Soner ≈ delta-tolerance > Leland > plain Black-Scholes.

**The line that decides what you should build**

Zakamouline: "When $\lambda < 0.001$, our approximation strategy and both the asymptotic strategies produce practically identical results." NIFTY-futures $\lambda$ is about **1.3 bp** $= 0.00013$ — an order of magnitude below that threshold. **For hedging NIFTY delta with NIFTY futures, Zakamouline buys you nothing over plain Whalley–Wilmott.** Build WW and spend the saved weeks on realised volatility. The calculus flips only if you hedge with options.

### The other half: the cost of *not* hedging

Derman & Kamal (Goldman Sachs, 1999) give the risk leg in closed form. For $N$ rehedges, the standard deviation of replication P&L is approximately $0.886\,\sigma\,\text{Vega}/\sqrt{N}$; expressed in vol points, the one-standard-deviation replication error is about $\sigma/\sqrt{N}$. Their measured numbers: daily hedging of a one-month ATM option gives a P&L standard deviation of **16.3% of premium**; quadrupling the frequency halves it to 8.1%.

**Mean P&L is zero.** Discrete hedging adds noise, not bias. That is the correct null and most people get it wrong.

### The answer, computed for your instrument

Sepp's contribution is the objective function the other two are inputs to: expected P&L is independent of hedging frequency, P&L standard deviation falls as $1/\sqrt{N}$, but expected transaction costs grow as $\sqrt{N}$ — not $N$, because each rebalance is smaller. So **Sharpe is humped in** $N$ and there is an interior optimum.

Worked for a weekly NIFTY straddle: five sessions, 1,875 minutes, $\sigma = 13\%$, $\lambda = 1.3$ bp.

| Rebalance | $N$ | Derman noise (vol pts) | Leland cost (vol pts) |
|---|---|---|---|
| Daily | 5 | 5.80 | 0.16 |
| Hourly | 31 | 2.30 | 0.40 |
| 15-min | 125 | 1.16 | 0.80 |
| 5-min | 375 | 0.67 | 1.35 |
| 1-min | 1,875 | 0.30 | 2.90 |

**The trade-off bottoms out around 5 to 15 minutes and is flat between them.** Daily hedging is indefensible — it throws away 5.8 vol points of noise to save 1 vol point of cost. Minute hedging is indefensible in the other direction. Redo this with your own $\lambda$ and $\sigma$; note the two columns are not strictly additive (one is a standard deviation, one is a mean drag), so the table approximates Sepp's Sharpe rather than replacing it.

### Three things about the Indian cost structure nobody states

1. **The spread is irrelevant; the tax is the cost.** STT is roughly 77% of $\lambda$. The half-spread on NIFTY futures is 0.01 bp — a hundred times smaller. Every US paper on hedging costs is about spread and impact. In NIFTY it is about a statutory levy.
2. **STT is sell-side only, which violates the symmetric-cost assumption of all four band models.** Leland, Whalley–Wilmott, Barles–Soner and Zakamouline all derive a *symmetric* band around delta. With a one-sided cost the true no-trade region is asymmetric — the sell boundary should sit further from delta than the buy boundary. **Nobody has published this for the Indian market.** It is a clean, small, publishable extension and it takes days.
3. **The cube root means the cost input barely matters — and that is the point.** Band $\propto \lambda^{1/3}$: to halve the band you must cut costs eightfold; doubling $\lambda$ widens it only 26%. Total cost over the option's life scales as $\lambda^{2/3}$, so a doubling of costs costs you 59% more, not 100% — *provided you re-widen the band*. Keep the band fixed and you eat the full doubling. Given the 1 April 2026 STT rise, this is live: $\lambda$ roughly doubled and your band should widen about 29%.

### Your Black-Scholes delta is systematically wrong on NIFTY

Hull & White's minimum-variance delta is the correction: $\delta_{MV} = \delta_{BS} + \nu_{BS}\,\partial\mathbb{E}[\sigma_{\text{imp}}]/\partial S$, estimated empirically as $\delta_{MV} = \delta_{BS} + (\nu_{BS}/(S\sqrt{T}))(a + b\,\delta_{BS} + c\,\delta_{BS}^2)$, fitted by OLS separately for puts and calls on a rolling window.

On the S&P 500 over 1.3 million quotes, out of sample, this delivered variance reduction of **26% for calls and 23% for puts**, rising to **42% for the highest-strike OTM calls**. The correction was negative in every moneyness/maturity bucket — *under-hedge calls, over-hedge puts* — and it beat a daily-recalibrated SABR using 312 parameters against SABR's 100,000.

NIFTY has a steep, persistent negative skew and a strongly negative NIFTY/India VIX correlation, so the correction is at least as large as SPX's. Two things follow:

- **Nobody has published $a$, $b$, $c$ for NIFTY.** Novel, cheap, high value.
- Hull & White explicitly confined their test to maturities beyond 13 days, noting that including very short maturities worsens results because of large gammas near the money. **That exclusion removes almost the entire NIFTY weekly complex**, which is where all the volume is. So you cannot import their coefficients; you must estimate your own, conditioned on days to expiry — and because NSE now runs a weekly Tuesday cycle, the coefficient will almost certainly be a function of position in that cycle, a structure the SPX literature does not have.

### Expiry day: the formulas fail, and they fail loudly

As $\tau \to 0$, ATM gamma diverges, so the Whalley–Wilmott band widens as $\tau^{-1/3}$ — past the entire feasible delta range. **The model is not asking for infinite trading. It is telling you the position is unhedgeable and you should stop.** Away from the money, gamma goes to zero but Zakamouline's $H_0$ diverges. Leland degenerates regardless.

The practitioner override is not a better band. It is to stop treating expiry day as a hedging problem and treat it as a *sizing* problem:

1. **Cap dollar gamma ex ante** — decide the maximum you will carry into the final session at position-entry time, when you can still choose. No rebalancing rule rescues a position that is too big on Tuesday afternoon.
2. **Your futures hedge has structural basis risk on expiry day by construction.** Final settlement is the index close (and since August 2026, an auction print), not the futures. A delta hedge in futures cannot replicate it. This is precisely what SEBI's +2% expiry-day ELM is compensating for — the circular's stated rationale is the basis risk of the expiring contract.
3. Roll or close before the final session if gamma is material, or hedge gamma with a same-expiry option rather than delta with futures.
4. After the close you hold an unhedgeable position through the settlement window. Size for that, not around it.

### What minute bars can and cannot measure

NSE's continuous session is 375 minutes. Time-based rules — rebalance every $\Delta t \geq 1$ minute — are **exactly** measurable on minute bars.

**Band rules are not, and the bias is one-directional.** A delta band is touched *inside* a minute. Check only bar closes and you under-count crossings, hedge late, and overstate the strategy. Use bar high/low to detect the touch and assume a fill at the band level and you have assumed a resting limit order filled at the touch — zero adverse selection. That is the standard silent optimism in band backtests.

**The rule of thumb**

NIFTY per-minute return standard deviation is about $13\%/\sqrt{250 \times 375} \approx 4.2$ bp. If the band's half-width in spot terms, $h/(\Gamma S)$, is less than roughly twice that (about 8 bp), a typical minute contains multiple crossings and minute bars cannot resolve the strategy. Narrow bands are exactly what low $\lambda$ and high $\Gamma$ produce — i.e. exactly NIFTY weeklies near expiry. **Calibrate time-based rules on minute bars honestly; treat band-rule minute-bar results as an upper bound and validate the winner on ticks.**

## 5.2 Realised volatility from minute data

Your single largest new asset, and no round-one agent said how it should be measured.

### The decision, already made on evidence

Liu, Patton & Sheppard tested 420 estimators across 31 assets, five asset classes and eleven years of tick data, with proper multiple-testing correction. **Nothing significantly beats 5-minute realised variance** except for one asset. Four operational findings matter more than the headline:

- Subsampling helps at 5 and 15 minutes; it *hurts* at 1 minute and below.
- Transaction prices beat mid-quotes at low frequencies; quotes win only at one tick or one second.
- Tick-time sampling beats calendar-time at 5 and 15 minutes.
- **The line that matters most for you:** "more sophisticated realized measures generally perform significantly worse in non-US markets than in US markets." The realized-kernel and two-scale machinery is tuned to NYSE/CME microstructure. It has no demonstrated advantage on NSE and some demonstrated disadvantage outside the US.

### The diagnostic you must run first — 30 minutes of work

Andersen, Bollerslev, Diebold & Labys's volatility signature plot: average realised volatility against sampling interval. Two shapes with opposite meanings.

- **Liquid asset:** RV highest at the smallest interval, falling and stabilising. Cause is *negative* serial correlation from bid-ask bounce. Naive 1-minute RV is biased **up**.

- **Illiquid asset:** RV *lowest* at small intervals, rising. Cause is *positive* serial correlation from stale trading. Naive 1-minute RV is biased **down**.

**This decides your instrument choice, and the answer is not obvious**

The **NIFTY 50 index** is a weighted average of fifty stocks that do not print synchronously. At high frequency it shows the *illiquid* signature — positive autocorrelation from staleness — so 1-minute RV on the index is biased **downward**, opposite to the standard bid-ask-bounce story. The **front-month NIFTY future** is a single extremely liquid instrument and will show the liquid signature. **Compute RV on the future, not the index.** Draw both signature plots and look at them before writing another line of code. Note also that the originators of the tool recommended 20 minutes for their own data, not 5.

### The overnight gap, which is decisive for weeklies

Hansen & Lunde give the MSE-optimal linear combination $RV_{\text{day}} = \omega_1 r^2_{\text{overnight}} + \omega_2 RV_{\text{open-close}}$, and rule out the two things everyone does: **do not** simply add the squared overnight return, and **do not** scale open-to-close RV by a constant. Both are dominated. Their DJIA finding puts about 80% of daily volatility in the active session, with weights tilted heavily toward RV because a single squared overnight return is a far noisier estimator of its period than RV is of the session.

**The DJIA number does not transfer.** NIFTY's overnight window is 15:30 to 09:15 — **17.75 hours**, spanning the entire US session and GIFT Nifty's overnight trading, plus a pre-open auction. **Nobody has published the active/inactive variance ratio for NIFTY.** You must estimate it, and it is genuinely important: a weekly option prices seven calendar days of variance while you can only measure five sessions of 375 minutes. Get this wrong and the entire implied-versus-realised comparison is biased.

### NSE intraday seasonality is not the textbook U

The only free study measuring NIFTY intraday volatility on real minute data (1-minute futures, 2011–2018) reports hourly-scaled volatility:

| | 09:15–10 | 10–11 | 11–12 | 12–13 | 13–14 | 14–15 | 15–15:30 |
|---|---|---|---|---|---|---|---|
| Vol (%) | **0.350** | 0.238 | **0.225** | 0.246 | 0.285 | **0.327** | 0.280 |

Open-to-trough ratio is 1.56, a variance ratio of 2.4. But **the last half hour is lower than 14:00–15:00**. NSE's shape is a **J with a clipped right arm**, not a U — the afternoon peaks at 14:00–15:00 and volatility *falls* into the close. Import a US-calibrated U-shaped diurnal factor and you will over-deseasonalise exactly the window that sets the daily settlement price and the expiry settlement value.

Stated plainly: this source does *not* pass the filter. No cost model, not a practitioner, a student working paper in a non-refereed archive, and its methodology text describes converting per-minute to hourly volatility by multiplying by 60 — which is wrong (it must be $\sqrt{60}$), and the reported levels are only consistent with correct scaling, so the description and the arithmetic disagree. Use the *shape* as a prior to check your own estimate against, and re-estimate everything on your own data. An honest weak source flagged as weak beats a confident US import.

Three NSE-specific facts verified from primary exchange documents that no paper covers:

1. A **pre-open auction runs 09:00–09:08**, so the 09:15 open already absorbs part of the overnight gap. The first minute bar is a post-auction bar, not a gap bar. Do not treat 09:15 as the gap.
2. The **15:00–15:30 window is the daily settlement window**. Volume and volatility there are mechanically driven by settlement, not information.
3. On expiry day that same window determined final settlement for every expiring index option — and **moved from Thursday to Tuesday** for contracts expiring on or after 1 September 2025, then was **superseded by the Closing Auction Session** on 3 August 2026. Estimate the periodicity factor separately per regime. That is at least four regimes, arguably six.

### The recommended estimator, and the first project specified

**Recommendation**

**Subsampled 5-minute realised variance on the front-month NIFTY future, computed in tick time, on transaction prices, open-to-close, plus a Hansen–Lunde-weighted overnight term — with the sampling interval confirmed, not assumed, by your own volatility signature plot.**

The project, step by step:

1. **RV pipeline.** Front-month future, minute bars, aggregated to 5-minute in tick time, subsampled across the offset grids. Compute bipower variation and MedRV on the same grid for the jump decomposition.
2. **Deseasonalise** with a jump-robust periodicity estimator, fitted *separately per regime*. Use the deseasonalised series for spot-vol and jump detection; use raw RV for the daily aggregate.
3. **Overnight.** Estimate NIFTY's active/inactive split yourself, then combine with Hansen–Lunde MSE-optimal weights. Report the estimated split — it is a publishable number in its own right.
4. **Match the clocks.** Remaining implied variance at minute $t$ is $\sigma_{\text{imp}}(t)^2\tau(t)$ with $\tau$ in *calendar* years to settlement, because the option prices calendar time including weekends. Remaining realised variance is the sum over surviving minutes *plus every overnight to expiry*. Mismatch these two clocks and the whole study is noise.
5. **Weight by dollar gamma, not equally.** This is the step that makes it a trade rather than a statistic. Delta-hedged short-option P&L is $\frac{1}{2}\int \Gamma_s S_s^2(\sigma_{\text{imp}}^2 - \sigma_s^2)\,ds$ — a *dollar-gamma-weighted* average of the variance gap. A variance risk premium that lives in the low-gamma part of the option's life is not tradeable. Compute both the plain and the gamma-weighted VRP and report the difference; if they disagree, the plain one is a mirage.
6. **Decompose and stress.** Split RV into continuous and jump components, locate the jump minutes, and re-run excluding jump days. Compare implied against RV *including* jumps — that is what the hedged P&L actually eats — but report the jump-free version alongside. The gap between the two tells you whether you are harvesting a variance premium or writing crash insurance.

Toolchain note, and an honest negative. The complete, peer-reviewed, free implementation is the R package `highfrequency` — realised kernels, two-scale, pre-averaging, bipower, semivariance, Barndorff-Nielsen–Shephard and Lee–Mykland jump tests, and the intraday periodicity estimators you specifically need. **There is no Python equivalent of comparable completeness or peer review.** The closest is Kevin Sheppard's MATLAB MFE Toolbox, which is the code the Liu–Patton–Sheppard study actually used. If you are a Python shop you will be writing RV estimators yourself or calling R. Better to know that now.

## 5.3 Sizing and risk of ruin

### Why full Kelly is wrong for options

1. **Kelly is optimal for the distribution you feed it, and your backtest does not contain your loss distribution.** A short-gamma book's P&L is a series of small positive numbers punctuated by a large negative one. With zero or one tail events in sample, $\mu$ is biased up and $\sigma^2$ biased down, so $f^* = \mu/\sigma^2$ is biased up *multiplicatively in both terms*. This is the dominant error, not a small one.
2. **The moment correction points the same way.** Sinclair & Brooks derive closed-form skewness and kurtosis of European option *returns* and show the higher moments shift the Kelly fraction materially — and for short options they shift it *down*. Flagged: SSRN blocks programmatic access, so the abstract and affiliation were read but not the 13-page body, and no formula from it is reproduced here. It is free and short; download it yourself.
3. **Kelly's own properties are bad.** Full Kelly's expected maximum drawdown is unbounded in the limit, and Thorp's own "when will Kelly dominate" analysis shows the horizon is long enough that a solo trader will be stopped out — by his own nerve or his broker — long before the asymptotics arrive.

### The fix is not "use half-Kelly"

Busseti, Ryu & Boyd's risk-constrained Kelly makes the drawdown tolerance the *primitive* rather than a fudge factor. Maximise $\mathbb{E}\log(r^\top b)$ subject to the usual constraints plus $\mathbb{E}(r^\top b)^{-\lambda} \le 1$, with $\lambda =$

$\log \beta / \log \alpha$ where $\alpha$ is the drawdown level and $\beta$ the tolerated probability.

It is a **convex** problem, its entire specification is about six lines of CVXPY, and the code is published. Its properties:

- The constraint bounds the *entire* CDF of minimum wealth: $P(W_{\min} < \alpha) \leq \alpha^{\lambda}$ for all $\alpha$, not just the one you specified. Their example: $\alpha = 0.7$, $\beta = 0.1$ gives $\lambda = 6.46$, so the probability of ever losing half your capital is bounded by $0.5^{6.46} \approx 1.1\%$.
- $\lambda = 0$ recovers unconstrained Kelly exactly; $\lambda$ is a risk-aversion dial.
- It *dominates* fractional Kelly in simulation — higher growth at the same drawdown risk, or lower risk at the same growth.

**The caveat that applies to all of this:** the method takes the return distribution as given. Feed it the empirical distribution of your backtested short-gamma trades and it will hand you a confidently oversized position, because the empirical distribution does not contain the loss that has not happened yet.

### Margin expansion is the constraint that binds first

This has nothing to do with the maths and will kill you before the maths does. SPAN risk parameters are revised **six times a day** — 11:00, 12:30, 14:00, 15:30, 17:00, end of day and beginning of day. **Your margin requirement can be raised four times during the session you are losing money in.** That is the documented, scheduled, public mechanism of margin expansion under stress. Add the +2% expiry-day ELM, which raises margin on the day your gamma is largest, and the removal of calendar-spread benefit on expiry day, and the ₹15 lakh minimum contract size, which quantises any Kelly fraction below a certain account size.

**The one-line rule**

Size every short-gamma position so that a single overnight gap of the worst magnitude you can *imagine* — not the worst in your sample — combined with a SPAN scan-range widening and the expiry-day ELM add-on, still leaves you funded; *then* take the risk-constrained-Kelly fraction for your stated drawdown tolerance and halve it, because your edge is estimated with far more error than your loss.

The halving is not superstition. It is a crude stand-in for the estimation variance of $f^*$, and you should replace it with the actual confidence interval once you have read Sinclair's paper on confidence intervals for the Kelly criterion.

## 5.4 P&L attribution: how you find out whether any of this works

### The free corpus, and a name correction

**Timothy Klassen, not Kevin**

PhD particle physics, then equity derivatives quant under Emanuel Derman at Goldman, then **designed the 2003 "new VIX" for Cboe**, then built Getco's options analytics infrastructure and quant team, then founded Vola Dynamics in 2016 with Jim Gatheral as an advisor. He is the single richest free source in this entire corpus for spot-vol dynamics and P&L attribution, and round one missed him by searching for "PnL attribution" as an academic topic rather than as a desk engineering problem.

His *Spot-vol Dynamics and Deltas for SPX Options* (13 pages, free) is directly executable on your minute data. It gives the estimation recipe verbatim: "the (realized) vol sensitivities can be determined reasonably well using, say, one day's worth of 1-minute or 5-minute snapshot data using a linear regression, with some outlier treatment for robustness." That is a one-day project for you.

Concrete magnitudes of the error you are currently making: ATM delta adjustment about $-0.06$; put-wing peak about $-0.08$; and in the call wing the correct delta is essentially *zero* while the naive Black-Scholes delta is still 0.03 — a 100% relative error on tradable OTM calls.

And the explicit falsification, which is the part to internalise:

> "Sticky-strike scenarios are still commonly used by risk departments. In equities, at least, neither [sticky-strike nor sticky-delta] has happened for 20y+. Sticky-delta implies SSR=0 for all terms. Never happens. Sticky-strike implies SSR=1. Even when vols are sticky-by-strike around ATM, they never are in the wings."

Typical skew-stickiness ratio is about 1.3, rising toward 2.0 on large down days, with **KOSPI often above 2 —** the one named non-US index, and plausibly a better prior for NIFTY than SPX is. Note also that SSR now has a term structure and is horizon-dependent: estimate it from 1-minute NIFTY data and you will get a different number than from close-to-close.

The free video series is the curriculum, in this order: *PnL Attribution #1: The Greeks → #2: Scenarios* (full revaluation versus greek attribution — i.e. where the unexplained residual comes from) → *Thetas and Time Conventions → Sticky-by-Strike or Sticky-by-Delta? Neither! → If sticky strike is not the answer, what is?* Then read the Spot-vol Dynamics PDF with a notebook open.

Caveat, stated because it is load-bearing: this is marketing collateral for a paid C++ library, and the self-serving claim is explicit — "good vol curves that can actually match the market implied vols for liquid names (none in the public domain do)". Since you built SVI, treat that as a hypothesis to test. His slides also argue SVI has insufficient shape flexibility and that SPX needs 16–18 parameters, with an empirical scaling law of roughly $n_{\text{params}} \approx (n_{\text{options}}/5)^{1/3}$. Note what that law implies: **NIFTY chains are far smaller than SPX's, so NIFTY needs materially fewer parameters — possibly making SVI adequate where it is not for SPX.** Test it; do not assume either way.

## The regulator already answered "how much unexplained P&L is too much"

BCBS d457's P&L Attribution test is the only quantitative, published, universally applied standard for whether a risk model explains its P&L. Compare risk-model P&L against hypothetical P&L (yesterday's positions revalued with today's market data, no intraday trading, no new deals) on a rolling 250-day window, using two statistics: **Spearman rank correlation** and the **Kolmogorov–Smirnov** statistic.

| Zone | Correlation | KS statistic |
|---|---|---|
| Green | $> 0.80$ | and $< 0.09$ |
| Amber | between — carries a capital surcharge | |
| Red | $< 0.70$ | or $> 0.12$ |

The footnote that answers your theta question directly: P&L due to the passage of time must be included and *treated consistently* across both measures. The regulator's answer to calendar-versus-business time is **pick one and be consistent —** inconsistency there is itself a model failure.

The instructive negative: Basel's own vega bucketing for equities is defined along *one* dimension, option maturity, mapped to 0.5, 1, 3, 5 and 10 years. It does **not** bucket equity vega by strike or moneyness at all; skew risk is pushed into a curvature charge. So the regulator's scheme is deliberately cruder than your book needs. Use it as a *floor* — a strike-bucketed vega grid is a strict improvement — and take the published maturity-decay correlation kernel as a defensible free starting point.

The honest limit: the PLA test compares model P&L to model P&L, not model P&L to realised cash. It answers "is my risk system consistent with my pricing system", which is weaker than "is my model right about the market". It is also distributional over 250 days, so it gives you *no per-day threshold* for "today's residual is too big". For that you must set your own bar, and nobody has published one.

## The Indian business-time problem, which is now a moving target

With a **Tuesday** expiry, the weekend sits *in the middle* of a weekly option's life rather than at the start. Under a calendar-day convention your Monday and Friday theta are the same number; under trading-day they are not, and the difference is the entire P&L of a short weekly in a holiday week. India runs 14–16 exchange holidays, unevenly clustered around October and November, so a Tuesday-expiry weekly spans seven calendar days but between **three and five trading days** depending on the week.

**Nobody has published on business-time theta calibration for NIFTY weeklies.** With minute data you are positioned to close it: estimate intraday versus overnight versus weekend variance shares directly from 1-minute returns, then compare the implied vol you back out under each time convention.

Honest negative worth knowing before you go looking: **there is no credible open-source P&L-explain engine.** The best-known hit is a single Jupyter notebook. Everything else is a zero-star toy repository, and every production implementation is

commercial and closed — Vola Dynamics, Murex, Bloomberg MARS. You will write this yourself, and that is a genuine, defensible piece of work.

# 6. Backtests that do not lie, and spending the minute-data advantage

## 6.1 The most instructive artefact in the entire corpus

**A peer-reviewed 0DTE result that died when its cost model was fixed**

Grigory Vilkov (Frankfurt School) is one of the few options academics who ships his replication code. In August 2026 he posted a correction notice on the repository, verbatim:

*"A transaction-cost unit-scale error caused the bid-ask half-spread to be charged at 1/100 of its true size (about 0.022 bp instead of 2.2 bp). The code is fixed in this commit, and the correction reverses the sign of the net-of-cost conclusions: no strategy or basket retains a positive net Sharpe ratio. Gross results are also affected."*

**One correction to how round one reported this.** The repository's `KNOWN-ISSUES` file states the result more carefully — "no structure retains a *materially* positive net Sharpe ratio" — and its own table does not support the stronger sentence in the banner:

| Structure | SR at mid | SR net (before) | SR net (after) |
|---|---|---|---|
| Put ratio spread | $+1.06$ | $+0.84$ | $\mathbf{-0.61}$ |
| Risk reversal | $+0.65$ | $+0.44$ | $\mathbf{+0.10}$ |
| Bear put spread | $+0.48$ | $+0.30$ | $\mathbf{-0.73}$ |
| Strangle / straddle | $-0.27$ | $-0.51$ | $-0.97$ |
| Iron butterfly / condor | $-0.56$ | $-0.96$ | $-2.67$ |

Risk reversal survives at $+0.10$. Use the careful wording, not the banner.

### The mechanism, exactly

The data panel mixed scales: bid-ask spread, mid and time value were *fractions* of spot, while return columns were *percent* of spot. The cost line subtracted a fraction from a percent, while the flat fee term was correctly scaled. Entry turnover for a two-leg strangle went from an apparent 0.24 bp to a true 24 bp.

### The part round one missed, and the most transferable lesson here

From the notice's own scope note:

> "Note that gross columns also move, because the binary target $y = \mathbb{1}\{\text{pnl_net} > 0\}$ depends on the cost. Correcting the cost relabels the training data, so the classifier learns a different — and weaker — signal. Anyone rescaling the cost only at the evaluation step, without also rebuilding the target, will get a more favorable number than the one reported here."

**If your cost model feeds your labels, fixing the cost model is not an evaluation-time patch.** You must retrain. This is the subtle version of the error and it will survive a casual audit.

**What actually changes for you**

A cost bug of exactly two orders of magnitude survived peer review, an SSRN posting and a public replication package, and was caught by an outside replicator reproducing a table to four decimals. **Treat every published 0DTE net Sharpe as unproven until you have read the cost line and checked its units.** Note the scale: 2.2 bp of half-spread was enough to invert every result on SPX. A NIFTY weekly wing spread is orders of magnitude larger than that.

## 6.2 How options backtests lie

In descending order of how much they catch, in options specifically:

1. **Re-price the whole backtest at three cost levels**: mid, mid $\pm$ 25% of half-spread, and full spread crossing. **If the sign of P&L changes, you have a cost-model result, not a strategy.** This is the Vilkov failure and it is the modal failure.

2. **Timestamp-shift test.** Move every entry and exit by $\pm 5$, $\pm 15$, $\pm 30$ minutes. A real edge degrades smoothly; a microstructure artefact or a look-ahead bug falls off a cliff. **Daily-bar backtests cannot run this test at all** — this is one thing your minute data buys immediately.
3. **Quote-staleness audit before any P&L is computed.** Per strike per minute, compute time since the quote last changed and spread as a percentage of mid. Recompute excluding any leg stale beyond $k$ minutes at decision time. **The delta between the two runs is the size of your lie.**
4. **Live-versus-backtest reconciliation on paper fills.** Run forward, log your intended price and the market's next print, and track the distribution of the gap. After about 200 fills you have an empirical slippage model that replaces every guess. Nothing else substitutes for this.
5. **PBO via combinatorially symmetric cross-validation** over your configuration grid, plus walk-forward with a refit cadence matching production — including the fit cost in the out-of-sample window.
6. **Assignment and settlement replay** as a separate unit test, not part of the strategy loop. Force expiry-day paths through the settlement rule and verify cash by hand for a few dates.

**On deflated Sharpe versus PBO — a blunt assessment**

The deflated Sharpe ratio is academically popular and practically half-useless, because it requires $N$, the number of trials, and **nobody honestly reports their own $N$ — including the people citing DSR**. **PBO via CSCV is the one to implement**: it splits the backtest into combinatorial train/test partitions and reports the probability that your in-sample-best configuration ranks below median out of sample. It needs no $N$ and is computable from artefacts you already have. Its limit: it catches *selection* lies, not *execution* lies. Both are live in options.

## 6.3 Open-source options backtesters: specific verdicts

**Trustworthy with work — QuantConnect LEAN.** A real event-driven engine with a reality-modelling layer and source-readable option pricing. But read their own docs first: the default fill model *does not consider size or volume*, and their greeks come from a QuantLib Black-Scholes that admittedly diverges from vendor implied vols. Out of the box it will fill you at prices no one would give you. **It has no meaningful Indian options support**, so for NIFTY you are writing your own engine regardless — use LEAN as a design reference for the reality-model separation, not as your runtime.

**Honest about itself, wrong resolution — optopsy.** Better than its reputation: it models commissions and several slippage schemes, and its README disclaims exactly the right things. But no margin model, no assignment model, chain-snapshot oriented rather than intraday, and the project is forked across three organisations — a reliable signal of abandonment. Screen structures on end-of-day chains with it; do not use it for anything at minute resolution.

**Actively misleading if used for options.** `vectorbt`, `backtesting.py` and `backtrader` with an option chain bolted on. None are options engines: vectorised frames make look-ahead free and silent, none carry a contract-lifecycle or assignment concept, and `backtrader` has no options support at all despite endless blog posts pretending otherwise.

**Closed-box Indian simulators** — AlgoTest, Sensibull, Opstra and the rest — have undisclosed fill assumptions and are product funnels. As evidence they are worth zero. As a *cross-check on your own chain reconstruction* (does my strike list on a given date match theirs?) they are mildly useful.

**Right philosophy, wrong asset class — `hftbacktest`.** Rust and Python, reconstructs the book from level-2 and level-3 feeds, models *feed latency and order latency separately*, and simulates fills using queue position. No options support — its value to you is as the reference implementation of what an honest fill simulator looks like, and as the tool for the crypto leg. It is also the concrete demonstration of why 1-minute options bars cannot answer passive-fill questions.

## 6.4 Verify what you actually received — this decides everything below

**Ask your data provider one question**

Is each 1-minute bar OHLC of the *last traded price*, or does it carry a *best bid / best ask snapshot*? Most Indian "1-minute options data" is last-traded-price only. If so, then for any strike outside the few liquid ones the "price" in a bar can be a trade from minutes ago, arbitrarily far from the live quote, and **every fill assumption you build on it is fiction**.
LTP-only minute data is still excellent for realised-variance and timing work and nearly useless for fill-sensitive work. Do not proceed until you know which you have.

### What is tractable at one minute, and what is not

**Tractable:** time-of-day realised-variance seasonality; intraday variance risk premium; intraday surface dynamics and per-minute refits; discrete delta-hedging error and gamma-scalping P&L; entry and exit timing sensitivity; expiry-day pin behaviour; synthetic-forward versus futures basis; event studies around the open and the settlement window.

**Not tractable — needs level-1 with timestamps at minimum, level-2 properly:** passive fill probability, queue position, adverse selection on resting orders, effective versus quoted spread, order-book imbalance signals, market making, and anything holding less than roughly 10–15 minutes.

**The working rule**

At 1-minute resolution, only backtest strategies that **cross the spread on entry and exit and hold at least 15 minutes**. Everything else is a guess wearing a Sharpe ratio.

## 6.5 Storage: pick this and stop thinking about it

Full NIFTY and Sensex option chains at 1-minute for several years is **tens of gigabytes compressed**, not terabytes. That is small, and it settles the question.

- **Hive-partitioned Parquet on local disk, with DuckDB as the query engine.** Partition by date and underlying, sort within file by expiry, strike, right and timestamp, compress with ZSTD. DuckDB queries Parquet directly and keeps peak memory bounded. **Do not use DuckDB's native file as your system of record** — its on-disk format is not stable across versions. Parquet is truth.
- **ArcticDB** is real, Man Group-built, and its bitemporal versioning is a genuine argument given that vendor restatements of options data are common. But it is last-writer-wins with no transaction isolation, and versions cost storage. **Verdict: not worth it at your scale.** Adopt it only if you start receiving daily corrections you must audit.
- **kdb, ClickHouse, QuestDB, TimescaleDB: no.** They are right at tick-scale on a shared team feed. At single-researcher minute scale they are a second job, which is the thing you said you did not want.
- **The piece everyone skips and everyone gets burned by:** a separate contract-metadata table with `valid_from` and `valid_to` — lot size, tick size, expiry date, strike interval. **Never join on lot size as a constant.** See Section 3.5 for why.

## 6.6 Edges that only exist at minute resolution

This is where the advantage is actually spent. Six, roughly in order of value.

1. **Time-of-day-adjusted realised variance against remaining implied variance.** Daily data cannot tell you that 1-minute RV of 12% annualised is *high* at 12:30 and *low* at 09:20. Deseasonalise, compare against the option's implied variance over its *remaining life only*, and you have an intraday VRP signal. Specified fully in Section 5.2.5. Highest value.
2. **The intraday implied-vol diurnal pattern is an execution edge before it is a strategy.** Both calls and puts show lowest implied vol at the open with call implied vol drifting to a peak at the close. Even if you never trade it directionally, it means "sell premium" backtests that fill at the close are systematically flattered and "buy premium at open" backtests systematically hurt — a bias daily-bar

work cannot even detect, let alone remove.

3. **Timing decomposition of any existing daily strategy.** Take any published Indian short-premium or calendar result, re-run it at every minute of the day, and plot P&L against entry time. In almost every case the majority of the "edge" concentrates in two 15-minute windows — which tells you it is a microstructure artefact rather than a risk premium. **This is a falsification tool daily data physically cannot provide, and it will kill more of your ideas than anything else you build.**
4. **Honest gamma-scalping and hedge-error estimation.** The variance of short-gamma P&L depends almost entirely on hedge frequency, and daily-bar backtests silently assume one hedge per day, thereby overstating short-straddle Sharpe. With minute data you can compute the actual hedge-error distribution as a function of rebalance interval and price the trade-off. **The single most decision-relevant thing minute data gives an options book.**
5. **Per-minute surface refits** — the direct extension of work you have already done. Track the time constant of smile relaxation after a jump, intraday ATM-skew mean reversion, and *persistent* no-arbitrage violations. Violations lasting a minute or two are stale quotes, which gives you a free model-based staleness detector. Violations persisting longer are either a real calendar/butterfly trade or a data defect — and distinguishing them is exactly what is tradeable and what nobody with daily data can even look at. India's expiry-day chains, with the deepest short-dated liquidity in the world, are arguably the best dataset on earth for this.
6. **Synthetic forward versus futures, per minute.** The put-call-parity forward and the futures price should track. When they diverge, one of your three inputs is stale. A cheap continuous data-quality monitor, and occasionally an actual box-spread signal. Impossible at daily resolution because everything is synchronised at the close by construction.

## 6.7 Indian option-chain traps, specifically

- **Lot sizes are point-in-time facts, not constants** — see Section 3.5.
- **Expiry day changed twice** in recent history, and once was a cross-exchange *swap*. Any day-of-week feature, any expiry flag and any DTE calculation must be derived from a dated contract master, never from a weekday rule.
- **Illiquid wings quote garbage.** Enforce a hard filter — drop legs with spread above a threshold percentage of mid, or with zero volume in the bar — *before* the strategy sees the chain, and **report how many candidate trades the filter killed**. That number is a result in itself.
- **Forward, not spot.** Compute implied vol against the synthetic forward from put-call parity per minute, never against the index spot. India's index carry is non-trivial and dividends are lumpy; using spot puts a fake skew in your surface that your fit will faithfully reproduce and you will then trade.
- **Expiry-day settlement mechanics.** STT on exercised in-the-money options is charged on intrinsic value — a large, asymmetric, frequently omitted cost that specifically penalises held-to-expiry positions. Section 7.2.4 shows the direction of that incentive has now *reversed* relative to the folklore.

# 7. India: the rules, the arithmetic, and the margin

Round one computed the crypto tax arithmetic in detail and never stated the NSE F&O treatment at all, then implicitly compared venues across the two. Round two computed both halves from the statute and the exchange's own files. Everything in this section is derived from primary documents; the unverified residue is listed at the end.

## 7.1 What it costs to trade, per round trip

**One correction that changes the premise**

The Finance Act 2026 raised derivative STT again with effect from 1 April 2026: options **0.10% → 0.15%**, futures **0.02% → 0.05%**, exercise **0.125% → 0.15%**. Any cost model carrying the October 2024 rates — which is most of what is published, and was round one's basis — is one revision stale.

### Index options, per rupee of one-way premium

| Component | Rate | bp of one-w |
|---|---|---|
| STT, sell leg only | 0.15% of premium | |
| Exchange transaction charge + IPFT, both sides | ₹3,553/crore × 2 | |
| GST at 18% on exchange charges | — | |
| Stamp duty, buy leg *(unverified)* | 0.003% | |
| SEBI turnover fee *(unverified)* | about ₹10/crore × 2 | |
| **Regulatory total, round trip** | | **ab** |
| Brokerage | broker-specific; ₹20/order is the discount standard | |

Pre-April-2026 the same stack was about 18.7 bp. **The options round-trip floor rose 27%.**

### Index futures, per rupee of notional

STT 5.00 bp (sell leg only) + exchange 3.66 + GST 0.66 + stamp about 0.20 = **about 9.5 bp**, against about 6.5 bp before. **Futures round-trip cost rose 46%.**

**The consequence you actually need**

A short NIFTY weekly straddle collects ₹11,495 per lot on ₹15,71,417 of notional (real 28 August 2026 prices). **One full round trip of that notional in futures costs 9.5 bp × ₹15.71 lakh = ₹1,493 — 13.0% of the premium collected.** Delta-hedging that straddle with total hedge turnover of twice position notional over its life burns **26% of the premium**. Three round trips of notional and the trade is a coin flip before you have priced a single greek.

**The Finance Act 2026 futures STT increase is, functionally, a tax on delta-hedging.** Combine this with Section 5: hedging cadence is now the single largest controllable cost line in a short-premium book, and the band should widen by roughly $(2.8/1.3)^{1/3} \approx 1.29\times$.

## 7.2 Tax: the F&O regime, from the statute

Section numbers are **Income-tax Act, 2025**, in force 1 April 2026. The 1961 equivalents are given once in brackets, because most published material still cites those.

### Non-speculative status

s.66(31) [old 43(5)] excludes "a specified derivative transaction" from the definition of a speculative transaction. s.66(33) defines that as trading in SCRA derivatives, carried out through a SEBI-registered broker electronically on a recognised exchange, supported by a **time-stamped contract note stating the unique client code and PAN**.

Note what is *not* required: unlike the commodity limb of the same clause, the equity-derivative limb

does not condition non-speculative status on the transaction being chargeable to STT. Screen-based, through a broker, with a compliant contract note — that is the whole test.

**Consequence:** NIFTY futures and options are **business income**, neither speculation business nor capital gains. Intraday cash equity, by contrast, *is* speculative and lives in a worse ring-fenced regime.

### Deductibility and set-off — the structural advantage

- **s.34(1)** [37(1)]: brokerage, exchange charges, GST on those, data feeds, VPS, terminal charges, internet, depreciation on hardware, interest on borrowed trading capital — all deductible.
- **s.32(1)(k)** [36(1)(xv)]: **STT itself is deductible**, provided the income is under business income. By contrast s.72(3)(b) expressly bars STT deduction when computing capital gains — so the moment you are classified as an investor rather than a business, your largest cost stops being deductible.
- **s.108**: loss from one source nets against any other source under the same head — a losing NIFTY book nets against a winning BANKNIFTY book automatically.
- **s.109(1)**: business loss sets off against *any other head including capital gains* — **except** s.109(1)(a), which bars set-off against **salary**.
- **s.112**: unabsorbed business loss carries forward **eight years**, against business profits only.
- **s.121**: a loss not determined in a return filed by the due date **cannot be carried forward at all**. File on time or the loss is gone permanently.

**The salary carve-out is your live constraint.** A salaried quant with an F&O book cannot use trading losses against salary in the loss year. They can be used against interest, rent, capital gains or other business income; otherwise they wait up to eight years.

### Turnover, audit, and the presumptive-taxation trap

Derivative *turnover* for audit purposes is the sum of favourable and unfavourable differences on squared-off trades **plus the premium received on sale of options**. That matters enormously to a seller: a book selling eight NIFTY straddles a week writes about ₹47.8 lakh of premium a year before a single rupee of P&L is counted. Scale it threefold and you are at ₹1.5 crore of "turnover" on a ₹25 lakh account.

The audit threshold is ₹1 crore, **substituted to ₹10 crore** where cash receipts and cash payments are each at most 5% of the total. A trader whose entire flow is bank-to-broker satisfies both, so **the effective threshold is ₹10 crore**.

**Why this might waste your time**

**Do not touch presumptive taxation (s.58, old 44AD).** Turnover includes written premium, which bears no relation to profit — so declaring the deemed 6% of a ₹2 crore turnover means declaring ₹12 lakh of income on a book that may have lost money. Decline to declare it in a later year, as any honest loss year forces, and s.58(7) locks you out of the section for **five years** while s.58(8) makes books and audit compulsory. It is a one-way door.

### The STT folklore is now false, and the incentive has reversed

Three regimes, from the primary record:

1. **Before 1 September 2019**: exercise STT was levied on the *settlement price* — the full notional. A NIFTY call ₹50 in the money attracted STT of roughly ₹1,970 on a contract worth ₹3,250. That is the massacre everyone remembers, and it was real.
2. **1 September 2019 to 31 March 2026**: the base changed to **intrinsic value**. The base collapsed by roughly 500-fold. A rate gap remained — exercise 0.125% versus sale 0.10% — so letting an ITM long expire still cost more STT than selling it.
3. **From 1 April 2026**: **both entries are 0.15%**.

**The arithmetic that inverts the standard advice**

At expiry an ITM option's market price is at least its intrinsic value. So for the holder of a long ITM option: selling it costs $0.15\%\times$ *premium* (intrinsic plus residual time value); letting it be exercised costs $0.15\%\times$ *intrinsic*. **Exercise is now weakly cheaper than sale, and strictly cheaper whenever any time value remains.**

The "always square off before expiry to dodge exercise STT" rule is a 2019-era artefact. What still argues for squaring off is settlement-price risk and, for stock derivatives, the delivery-margin ladder — not STT.

**The asymmetry that survives:** exercise STT falls on the *purchaser*, who paid no STT on entry. A long ITM position therefore carries a 15 bp exit tax on intrinsic that a naive P&L model omits entirely. On a synthetic long built from a deep ITM call — long the 22,000 strike with NIFTY at 24,176, intrinsic 2,176 — that is **₹212 per lot, every expiry, forever**, on a position whose premium is a small fraction of notional.

### Advance tax turns a volatile P&L into a working-capital problem

Advance tax is payable if the liability is ₹10,000 or more, and F&O business income has no withholding against it, so effectively the whole liability must be prepaid: at least 15% by 15 June, 45% by 15 September, 75% by 15 December, 100% by 15 March. Shortfalls carry interest.

**The structural problem nobody states:** you must estimate the whole year's P&L by 15 June to get the first instalment right. A book that makes its money in Q4 overpays; a book that gives back Q1 profits in Q4 has prepaid tax on income that no longer exists and gets it back as a refund a year later. The salary carve-out compounds it — you cannot even net a Q3 trading loss against salary already taxed at source.

## 7.3 The two regimes side by side

Same strategy, same gross edge, same turnover. This is the comparison round one never made.

### Case 1 — a low-turnover short-vol book (the favourable case for crypto)

Real inputs: NIFTY 24,175.65 on 28 August 2026, lot 65, notional per lot ₹15,71,417, one 24,200 weekly straddle collecting ₹11,495. Book: ₹25 lakh equity, eight lots, one cycle per week, 52 cycles. Outcome shape 72% win at $+$₹40,000 and 28% lose at $-$₹63,000, giving mean ₹11,160, standard deviation ₹46,247, and $r = 0.241$ — a genuinely good short-vol book with annual Sharpe 1.74.

| | NSE F&O | VDA regime |
|---|---|---|
| Gross annual edge | ₹5,80,320 | ₹5,80,320 |
| Tax base | net of year, after costs | ₹14,97,600 (winners only) |
| Tax at 31.2% | ₹1,81,060 | ₹4,67,251 |
| **Net** | **₹3,99,260 (16.0%)** | **₹1,13,069 (4.5%)** |
| Withholding in-year | — | ₹47,819 = 42% of the year's net profit |

Same strategy, same edge, same turnover: **the crypto regime keeps 28% of what the F&O regime keeps** — and this is the *flattering* case, because turnover is only 1.9 times equity.

### Case 2 — a high-turnover delta-neutral book (where it breaks)

Same ₹25 lakh equity, 5,000 round trips a year, ₹5 lakh notional each — ₹250 crore of annual turnover, a thousand times equity. Per-trade edge 2 bp of notional against per-trade volatility of 40 bp, so $r = 0.050$. Gross edge ₹5,00,000.

| | NSE F&O | VDA regime |
|---|---|---|
| Gross edge | ₹5,00,000 | ₹5,00,000 |
| Regulatory costs | −₹71,100, **deductible** | −₹71,100, **not deductible** |
| Tax | ₹1,33,817 | **₹13,24,255** |
| **Net before withholding** | **+₹2,95,083 (+11.8%)** | **−₹8,24,255 (−33.0%)** |
| 1% withholding on transfers | — | ₹2.5 crore = **ten times the entire equity** |

The tax alone converts a +20% gross book into a −33% net book. The withholding makes it unfundable before the tax question is even reached.

## 7.4 Margin as a simulable object

Named as a requirement in round one and never sourced. Round two implemented the model from published parameters and public daily files.

### The specification, exactly

**PRISM** is NSE Clearing's real-time position monitoring and disablement layer; **SPAN** is the margin computation inside it.

- **Price Scan Range**, index derivatives: six standard deviations scaled by $\sqrt{2}$ (the two-day margin-period-of-risk scaling), **subject to a floor of 9.3%** of the underlying.
- **Volatility Scan Range**, index: 25% of annualised EWMA volatility, **floor 4%**.
- **Sixteen scenarios**: price moves of $0, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm 1$ times PSR crossed with volatility up and down, all at 100% weight, plus $\pm 2\times$ PSR at **35% weight**. Worst-scenario loss is the maximum across all sixteen.
- **Extreme Loss Margin** is a second, *non-portfolio* layer: 2% of notional on short index options, 3% deep OTM, 5% beyond nine months, plus **an additional 2% on options expiry day**. Notional for a short option is the value of the underlying index, **not the premium**.
- **Calendar spread charge**: 1.75% of the far-month contract for index derivatives, charged *in addition* to worst-scenario loss — and the benefit is **withdrawn entirely on expiry day**.

**The structural fact for a spread trader**

SPAN is portfolio-based. **ELM is not.** It is levied on short option contracts and *long options do not reduce it*. This is why buying wings collapses your SPAN and barely touches your total margin.

### The numbers, computed from public files

Inputs, all from NSE's own files for 28 August 2026: underlying 24,175.65, lot 65, notional per lot ₹15,71,417; daily volatility 0.847876%, annualised 16.1986%; $6\sigma\sqrt{2} = 7.19\%$, so **the 9.3% floor binds**, giving PSR = 2,248.3 points and VSR = 4.05 vol points.

| Position (1 lot) | Premium | SPAN WSL | ELM | **Initial margin** | % notional |
|---|---|---|---|---|---|
| Weekly ATM short straddle, 4 DTE | 11,495 | 1,36,769 | 62,857 | **1,99,625** | 12.70 |
| Weekly $\sim 1\sigma$ short strangle, 4 DTE | 718 | 1,21,546 | 62,857 | **1,84,402** | 11.73 |
| Monthly ATM short straddle, 32 DTE | 36,189 | 1,19,923 | 62,857 | **1,82,780** | 11.63 |
| Monthly short strangle, 32 DTE | 3,916 | 89,198 | 62,857 | **1,52,055** | 9.68 |
| **0DTE ATM short straddle** | 11,495 | 1,36,769 | **1,25,713** | **2,62,482** | **16.70** |
| 0DTE short strangle | 718 | 1,21,546 | 1,25,713 | **2,47,259** | 15.73 |
| Iron condor (straddle vs wings) | 11,008 | **21,492** | 62,857 | **84,349** | 5.37 |

**Verdict on round one's figures.** The ₹1.5–2 lakh per lot for a short strangle is **correct** — but for a different reason than supposed: the lot is now **65, not 75**. The ₹3–3.25 lakh for a 0DTE short straddle is **about 15% too high**; the current figure is ₹2.62 lakh. The old number was right at lot 75.

### Three non-obvious results worth keeping

1. **SPAN margin for a short option is almost time-insensitive.** The 0DTE and 4-DTE straddles show the *same* worst-scenario loss. With a 9.3% price scan, intrinsic dominates regardless of time to expiry. **The entire 0DTE margin uplift — 24% — is the expiry-day ELM, nothing else.**
2. **Buying wings collapses SPAN by 84% and total margin by only 58%**, because ELM does not net across contracts. The iron condor's margin is 75% ELM. **Any margin-efficiency work on defined-risk structures must optimise against ELM, not against SPAN.**
3. **The 9.3% PSR floor was binding** ($6\sigma\sqrt{2} = 7.19\%$), and has been for most of recent years. **In calm markets Indian index option margin is not a VaR number at all — it is a flat 9.3% plus 2% administrative floor.** Margin therefore does *not* fall when volatility falls, which is precisely when a short-premium book wants leverage. Model it as a floor, not as a risk measure.

### The fact that makes a margin-aware backtest feasible

**Beginning-of-day parameters**

NSE Clearing's own risk FAQ: "The computation of the margin requirement of SPAN+ELM (both intra-day and EOD) … shall be based on the **fixed Beginning of Day (BOD) margin parameters**." **Your intraday margin path is fully reconstructible from the BOD parameters plus the intraday price path.** You do not need the intraday SPAN files — you need one parameter set per day and a tick series.

And you must simulate **peak** margin, not end-of-day margin. Clearing corporations take a minimum of four random client-wise snapshots per day, and the higher of the end-of-day and peak shortfalls is penalised, at 0.5% or 1.0% per client per segment per day. **A strategy that ends the day flat but peaks at three times margin mid-session has a real, penalised requirement it never shows on the closing book.**

**Forced liquidation is an event, not a slippage haircut.** Warnings fire at 70%, 80% and 90% utilisation; **at 100% the trading facility is withdrawn** — not a margin call, the terminal stops — and on failure to restore, NSE Clearing may close out positions by placing counter orders *without any notice*. Simulate it as a hard state transition: no new orders, and existing positions liquidated at market by a counterparty who does not care about your exit price. For a short-gamma book this is the mechanism by which a four-sigma day becomes a twelve-sigma P&L.

### Other components a backtest must carry

- **Margin on consolidated crystallised obligation** — net payable across premium and futures mark-to-market, blocked as margin.
- **Delivery margin** for stock derivatives, laddered from four days before expiry: 10%, 25%, 45%, 70%. **Index derivatives are cash-settled and escape this** — which is exactly what kills dispersion (Section 7.6).
- **Additional margin on extreme scenarios** — a second, wider stress test at a 20% market fall and 17.74% rise, sitting behind the 9.3% PSR and targeting precisely the naked short-option book.
- **Cash collection of option value** on deep-OTM shorts — requires *cash*, not collateral.
- **At least 50% of liquid assets must be cash or cash equivalents.** To carry a ₹1.84 lakh short strangle you must hold at least ₹92,000 in cash or T-bills earning a T-bill yield at best. The true funding cost of an Indian options book is therefore roughly half a repo rate plus half the opportunity cost of a haircut-reduced pledge, and that belongs in your cost model alongside the 23.7 bp. Verified at clearing-member level in the SEBI Master Circular; the identical requirement is applied down to the client by brokers, but the primary circular imposing it at client level was not reached.

**You do not need the SPAN risk-parameter file, and you cannot get it**

The `.spn` file is distributed to members via the NSE Clearing extranet; every anonymous download path returns 404, and the page titled "SPAN risk parameter files" contains only the parameter table with no file links. There is also **no public historical archive**, so you cannot backtest historical margin expansion from primary files.

**The good news:** everything that matters for index derivatives is either a published constant (9.3% PSR floor, 25%/4% VSR, 1.75% calendar spread, 2%/3%/5% ELM plus the 2% expiry-day add-on) or in a public daily file — the volatility file for $\sigma$, the market-lot file for lot size, the bhavcopy for prices. **The entire table above was produced from public files and published parameters, and it is about sixty lines of Python.** Expect single-digit percentage deviation from the official number, not orders of magnitude. **No public, runnable Indian SPAN implementation exists** — this is a genuinely publishable artefact and a better use of a week than reading another paper.

## 7.5 Indian code and craft: the honest finding

**Verdict: essentially nobody clears the bar for Indian options**

A systematic GitHub search across every plausible query, reading the actual cost code in each candidate, found three populations and none of them is what you want: broker-API wrappers and data libraries (real and useful, but plumbing with no cost model); live order-placement scripts for short straddles (the bulk of it, with no backtester behind them at all); and recently vibe-coded "platforms", mostly pushed in the last six months, near-zero stars, README claims far ahead of the code.

Searching "nifty implied volatility surface" returns 375 repositories, *none* of which is a serious NIFTY vol-surface implementation. Searching "dispersion trading implied correlation" returns *zero* India-relevant repositories.

**For someone who has already implemented SVI and SSVI, there is no Indian open-source prior art to stand on. You are not behind the field; there is no field.**

What that implies practically: the Indian options-research commons is a *data* commons, not a *methods* commons. Your comparative advantage against the domestic retail population is not the model — it is having a correct cost and settlement model at all. Treat US and European sources as the methods library and Indian sources as the data-and-rules library, and do not expect to find the two joined anywhere.

### An audit, because "realistic Indian cost model" was a claim worth checking

Round one's social sweep flagged one repository as "the single most on-target" Indian item, on the strength of a README claiming a realistic Indian cost model. Round two cloned it and read the code. **The claim does not hold.** Four defects, two disqualifying:

1. **The STT rate is pre-October-2024**, hardcoded at 0.0625% with a docstring claiming currency "as of 2025". The bundled sample data spans July 2025 to June 2026 — entirely inside the post-change regime. For a short-premium strategy STT is the largest cost line, and this understates it by 37.5% of its true value *in the favourable direction*.
2. **The expiry/exercise path is dead code.** The exercise flag is never passed anywhere in the repository, and the engine squares off unconditionally at 15:15 — so the single most-traded Indian structure, an expiry-day short straddle carried into settlement, is not backtestable at all. The flag is also on the wrong side: exercise STT is paid by the *buyer*, but the code only reaches that branch on sells.
3. **Lot sizes are a static snapshot with no time dimension**, and two are wrong against the very circular the file cites.
4. **Slippage is one tick, flat** — which is 0.025% on a ₹200 ATM option and 2.5% on a ₹2 wing. No spread model, no depth, no queue, and fills occur at the close of the same bar the decision was made on.

**Salvage:** the architecture is decent and the historical expired-chain integration is genuinely valuable

— expired option chains without a paid vendor. Fork it for the data layer, throw away the cost model, and never use the shipped sample results.

### The two Indian sources that do survive

**A first-person cost post-mortem.** An independent write-up documenting a signal with a monthly Spearman information coefficient of 0.1758 ($n = 106, t \approx 12.4$) — genuinely predictive — that nonetheless returned **−7.15% a year with Sharpe −3.27** over 2015–2024, from 541 trades a year at about 0.35% round trip. The cost stack is modelled from first principles including Almgren-Chriss market impact and stock-lending cost, not a flat basis-point assumption, and the limitations section names survivorship bias as the single most important caveat and flags that the short leg ignores the F&O ban list. Passes tests (2), (3) and (4). *Caveat: the code is not published, so it fails (1), and it is cash equities rather than options. Read it for the cost discipline.*

**One correctly-built cost model in open source.** A small repository whose cost module gets delivery equity right and — critically — **puts every rate in a configuration file rather than in source**, precisely so a regulatory rate change does not silently invalidate old runs. That is the design decision the audited repository got wrong, and it is the one to copy. It also benchmarks against a *costed* buy-and-hold and publishes a table in which three of its four strategies lose to it, and says so. *Caveat: no options.*

**One genuinely useful instrument.** A live-trading framework that *records realised slippage* — market price at decision versus actual fill, per order type, to CSV. That is an empirical slippage distribution for Indian options rather than an assumption. There is no backtest cost model in it; use it as an instrument, not a backtester.

**And one academic thread worth pulling.** Kumar, Agarwalla, Varma & Virmani, *Harvesting the volatility smile in a large emerging market: A Dynamic Nelson–Siegel approach* (*Journal of Futures Markets* 43(11), 2023) models the NSE smile as a term structure *in delta*, estimates it by Kalman filter, sorts on model-implied mispricing and trades the extreme deciles. This is the closest published thing to "SVI/SSVI, but for NSE, evaluated as a trade." **It is paywalled and round two could not confirm whether it nets transaction costs** — which, given the strategy trades single-stock options monthly across deciles, is the question that decides whether the result survives. **Pull it via IIT KGP and read the cost treatment first.** This is the single most important open question in the Indian half of this report.

## 7.6 Dispersion in India: the measured verdict

Computed from the actual NSE F&O bhavcopy for 27 August 2026, cross-referenced against the live NIFTY 50 constituent list.

| | Underlyings | Series listed | Series with a trade | Notional turnover |
|---|---|---|---|---|
| Stock options | 210 | 23,741 | 7,893 (33%) | ₹3.25 lakh crore |
| Index options | 6 | 4,700 | 1,083 (23%) | **₹77.3 lakh crore** |

**Index options are twenty-four times the entire single-stock option complex.** For the NIFTY 50 constituents specifically: all fifty have listed options, so on a naive screen the trade looks feasible. But aggregate constituent option notional is ₹1.41 lakh crore against NIFTY index options at ₹74.9 lakh crore — a **53:1** ratio. Median per constituent is ₹1,789 crore of notional across 46 traded strikes, and **eleven of fifty are below ₹1,000 crore a day**, the weakest at ₹214 crore across 36 traded strikes.

**Verdict: not viable — and liquidity is the *least* of the four problems**

1. **Single-stock options are physically settled.** Index options are cash-settled; stock options are not. A long constituent leg held into expiry generates a delivery obligation of *full notional* on every ITM leg, with delivery margins ramping from four days out. Across fifty legs this is not a margin inconvenience — it is a different business.
2. **The expiry calendars no longer line up.** Since 1 September 2025 single-stock options are **monthly only**. NIFTY has weeklies. The natural weekly dispersion trade has no stock leg at all, and the monthly trade puts your entire short-index leg into the one expiry where the +2% ELM and the removed calendar-spread benefit both bite.
3. **Margin has no correlation offset across underlyings.** SPAN scans each underlying independently. The whole economic point of dispersion — that the basket is nearly hedged — earns you nothing.
4. **Eligibility churn kills the backtest even where it spares the trade.** See below.

**And the ratio points the wrong way anyway.** Published work on Indian single-stock options finds the embedded risk premium is *higher* than in other markets and higher than theory suggests — implied volatility consistently exceeds realised "by too large an amount to be easily explained as a rational risk premium." That is the wrong sign for classic dispersion, where you want to *buy* cheap constituent vol. If anything the Indian data argues for *reverse* dispersion — short constituent vol, long index vol — which is exactly the leg physical settlement punishes hardest.

If you still want correlation exposure, BANKNIFTY against its twelve constituents is the one place the basket is small enough and every name is top-decile liquid. Both legs are now monthly-only, which at least makes the calendars match — at the cost of one observation per month.

### Why the single-stock universe is non-stationary

Eligibility is determined by a *rolling six-month window on cash-market statistics*, re-reviewed continuously. In August 2024 the thresholds were tripled in one step: median quarter-sigma order size ₹25 lakh → **₹75 lakh**, market-wide position limit ₹500 crore → **₹1,500 crore**, average daily delivery value ₹10 crore → **₹35 crore**. Fail any criterion for three consecutive months and the stock exits, with **no re-inclusion for one year**. A separate product-success framework adds turnover and open-interest floors.

**Any study that takes today's roughly 210-name F&O list and runs it backwards has survivorship bias of exactly the worst kind**: names were removed *because* their liquidity deteriorated. There is no free point-in-time constituent file — you must build one by differencing the daily bhavcopy universe, which you can do.

## 7.7 What could not be verified in the Indian material

Stated explicitly rather than glossed.

- **SEBI turnover fee** and **stamp duty** rates on derivatives — the regulations page serves its text via a JavaScript call that could not be resolved, and the Stamp Act schedule 404'd. Combined they are about 0.3 bp and move no conclusion.
- **GST at 18% on brokerage** — stated from general knowledge, not verified against the primary notification.
- **Whether NSE transaction charges are still at the September 2024 level** in August 2026. Check the circular archive before using the number.
- **The Income-tax Act 2025 "as amended by Finance Act 2026"** consolidated PDF is 403-blocked to non-browser clients. The gap was closed by reading the Finance Bill's amendment clauses directly and confirming none of the load-bearing sections were touched — but re-verify in a browser before relying on it.
- **Whether the November 2024 +2% expiry-day ELM has since been modified** by any later risk circular. It took effect; it was not traced forward.

- **Whether the DNS smile paper nets transaction costs.** The single most important open question in this section.

# 8. Crypto

Crypto was your top-weighted area, and it splits cleanly in two. As a **research venue** it is excellent and under-exploited: free tick data, real term structure, a risk-based margin engine at small account size, and a surface whose behaviour genuinely differs from equity index vol in ways nobody has published with costs. As a **P&L venue** for an Indian resident it is, on the arithmetic in Section 3.2, close to unusable for exactly the strategies you are training to build.

This section is what round two measured against live APIs on 28 August 2026. Every number below is reproducible with an unauthenticated request.

## 8.1 Strategy families: alive or decayed

### Perpetual funding carry — alive, but the risk is not where retail thinks

Funding has structurally compressed: 2021 bull-phase prints of 0.15–0.20% per eight hours (over 200% annualised) have given way to a 2024–2026 baseline near **0.01% per eight hours, about 11% annualised** before costs, spiking only in leverage-euphoria windows.

**The failure mode now has a date and a body count**

On 10–11 October 2025, roughly $19 bn of open interest was liquidated across 1.6 million accounts, $3.21 bn in a single minute. The mechanism that matters: **auto-deleveraging forcibly closed market makers' short perp legs against their long spot** — the exchange took away the profitable side of the delta-neutral book, leaving participants unhedged long into a 14% drawdown. A delta-neutral yield product marketing about 18% subsequently depegged more than 77%, with post-mortems tracing the damage to ADL breaking hedges.

**ADL is an uncompensated, unhedgeable forced exit that fires precisely at the tail**, and it ranks counterparties by profit times leverage — so being *more* profitable and *more* hedged puts you *earlier* in the queue. This has no analogue on NSE or the OCC. Whatever risk framework you carry over from the index work will not model it.

Capital is meaningful only above roughly $50–100k, because you run two collateral pools and must hold buffer against margin calls on both. Latency is low for entry and *high for the unwind* — October punished slow unwinds. A runnable multi-venue implementation exists in Hummingbot; read its open issue documenting a double fill on Hyperliquid after the strategy is stopped before you trust it. Your hedge is only neutral if the order state machine is correct.

The claim to disbelieve: a "19.26% annual, 0.80% max drawdown, positive all 12 months of 2025" figure circulates widely with no attribution and no cost model. A strategy that was positive in all twelve months of 2025 was, by construction, not honestly accounting for October.

### Market making — alive on CEX perps, folklore on AMMs

Solo market making is reachable on a small number of mid-cap perps, with a matched-region VPS. It is not reachable on BTC/ETH top-of-book against professional firms. Before assuming the economics, look at what a protocol's own professionally-run market-making vault earns *with* rebate and liquidation privileges you will not have: historical returns in the range of 3–8% annualised with negative months, and profit concentrated in a few high-volatility events. Your unlevered solo expectation should be *below* that, not above it.

**Why this might waste your time**

The AMM liquidity-provision version is mostly a losing trade dressed as yield. An LP is profitable only if fees exceed loss-versus-rebalancing, and one analysis of seventeen concentrated-liquidity pools found $199.3m of fees against $260.1m of impermanent loss — LPs were $60.8m worse off than simply holding. Concentrated liquidity is a short-gamma position sold to arbitrageurs at a price you do not set, and headline fee APR is gross of LVR. Skip unless you are specifically researching LVR mitigation.

### Cross-exchange and triangular arbitrage — decayed to folklore

No source passing the filter shows a solo retail trader profitably running latency arbitrage in 2025–2026. The arithmetic: cross-venue spreads sit roughly 0.1–2%, and at 0.1% taker fee per side you need 0.2% to break even — before slippage, before the fact that capturing it requires pre-positioned inventory on both venues, and before the funding cost of that idle split capital. The one legitimate residual is *funding-rate dispersion across venues*, which is slow-moving and capital-rewarded rather than speed-rewarded — and that is the carry trade, not this.

### Momentum — partly decayed, and the decay is regime-shaped

Time-series momentum produced negative annual returns through the choppy 2022–2023 regime, while cross-sectional momentum limited losses better because its market-neutral construction survives the absence of a trend. The effect is now regime-conditional rather than unconditional.

**Why this might waste your time**

Crypto time-series momentum is the most academically over-published and practically fragile item here. **Every backtest you will find applies no funding cost to the perp leg** — and holding a levered long perp through a positive-funding regime costs you precisely the 11% a year the carry trade is trying to earn. Momentum backtests that ignore funding are not backtests. Build the funding-inclusive version; it will look materially worse than every published result, and that gap is the honest answer.

## 8.2 Deribit market structure, measured

### The listing ladder is a step function, and that is the big difference

Deribit lists four dailies, three weeklies, three monthlies and four quarterlies, with strike intervals that step by tenor and moneyness — $500 at four days or less, $1,000–$5,000 at a week to a month, $2,000–$10,000 at three months. Measured ladders confirm the published policy exactly.

**This is the single biggest structural difference from NIFTY.** Deribit's strike spacing is a step function of tenor and moneyness, not a constant, and the far expiries carry strikes to four times spot. NIFTY gives you a dense uniform ladder over a narrow band; Deribit gives you a sparse non-uniform ladder over an enormous one. Your surface fitter cannot assume a uniform grid.

Open interest is extremely concentrated: the two big quarterly expiries carried **68.7% of all BTC open interest** at the time of measurement.

### The fee cap is the number with no NIFTY analogue

Options fees are **3 bp of underlying notional, capped at 12.5% of premium**, maker and taker alike — there is no maker rebate in options. Worked through at spot near $79,400:

| Premium | as % spot | Fee one way | % of premium | Round trip | Binding |
|---|---|---|---|---|---|
| $100 | 0.13% | $12.50 | 12.50% | **25.0%** | cap |
| $190 | 0.24% | $23.75 | 12.50% | **25.0%** | cap |
| $437 | 0.55% | $23.82 | 5.45% | 10.9% | notional |
| $1,826 | 2.30% | $23.82 | 1.30% | 2.6% | notional |
| $7,275 | 9.16% | $23.82 | 0.33% | 0.7% | notional |

The cap binds below a premium of **0.24% of spot**. Below that you pay a flat 12.5% each way. Above it you pay a fixed dollar amount per BTC *regardless of the option's value*.

**NSE charges on premium; Deribit charges on notional — and it inverts which part of the surface is tradeable**

Every OTM wing and every short-dated option on Deribit carries a fee that is enormous relative to its premium, while long-dated expensive options are nearly free. Any strategy that trades cheap options frequently — gamma scalping the dailies, wing buying, ratio structures — is dead on arrival at retail fee tier. This is the exact opposite of the NIFTY incentive.

### The tick is a second hidden tax

Deribit's inverse-option tick is a fixed *BTC* amount, so its dollar value scales with spot, and it steps up at higher prices. Measured across two-sided BTC options:

| Expiry | $n$ | Tick as % of premium | Quoted spread % | Median premium |
|---|---|---|---|---|
| Front daily | 41 | 2.41% | **16.87%** | $1,648 |
| Next daily | 48 | 2.80% | **16.58%** | $1,241 |
| ∼1 week | 61 | 2.08% | 6.67% | $1,866 |
| ∼1 month | 112 | 0.66% | 3.18% | $6,044 |
| ∼10 month | 100 | 0.33% | 2.06% | $12,148 |

Across all two-sided BTC options with premium under $500: **median tick is 9.09% of premium and median quoted spread is 24.00%**. Combine with the fee table: on a sub-$200 option you pay 25% round-trip in fees *plus* cross a roughly 24% spread. **Your edge must exceed roughly half the premium.** The Deribit wings and dailies are untradeable at retail size, and that is a fee-and-tick fact, not a liquidity opinion. Concentrate on the big quarterlies where open interest is and spreads are 2–3%.

## 8.3 The crypto surface behaves differently, and it is measurable in thirty lines

Live 25-delta skew term structure, computed from marks at spot $79,207 (BTC) and $2,505 (ETH):

| Expiry | Days | BTC ATM | BTC RR25 | BTC BF25 | ETH RR25 |
|---|---|---|---|---|---|
| Front daily | 0.7 | 33.1 | **+1.14** | +0.86 | **+2.82** |
| 1 week | 6.7 | 36.8 | **+2.44** | +0.76 | **+3.66** |
| 1 month | 27.7 | 37.5 | +0.26 | +1.05 | +1.11 |
| 2 months | 62.7 | 37.5 | +0.06 | +1.17 | +0.99 |
| 3 months | 90.7 | 39.4 | **−0.28** | +0.78 | +0.48 |
| 4 months | 118.7 | 40.3 | **−0.38** | +0.75 | −0.27 |
| 10 months | 300.7 | 42.5 | **−0.85** | +1.35 | +0.27 |

Two things an equity-index trader would not predict, both directly observable:

1. **The skew sign flips with tenor.** Short-dated BTC risk reversals are *positive* — calls bid over puts by one to 2.4 vol points — and go negative only beyond about ninety days. ETH is call-skewed nearly the whole curve. In SPX the risk reversal is negative at every tenor. **Any model you port from equities with a hard-coded negative correlation parameter will fight the front of this surface.**
2. **Term structure is in steep contango while the front is call-skewed** — cheap near vol, call-skewed, upward sloping. That combination is the crypto signature.

Butterflies are uniformly positive and remarkably stable at +0.7 to +1.4 across the whole curve for both coins, which is the one thing that does look like equities.

### The best answer to "does SVI calibrate here, and what broke"

Echenim, Gobet & Maurice (CNRS / École Polytechnique with Kaiko's quant data lead), free full PDF, using real Deribit BTC data. It contains:

- A **theorem** that calibrating SVI on mid-prices produces a *systematic underestimation* of implied volatility in the wings. Not a heuristic — a proof, with the direction of the bias.

- The reason: "the relative tick size (and thus the relative spread) for options on cryptos is roughly **10 to 50 times larger** than for options on traditional markets." The measurement above independently confirms it.
- An **anchor price** built from bid and ask whose bias is third-order in the tick size, replacing the mid.
- A data-augmentation procedure for one-sided and missing quotes — exactly the failure you will hit on Deribit wings.
- Hard liquidity counts: roughly **30 BTC option trades per hour across all strikes** for one-week and one-month expiries, about 5 per hour for three- and six-month, and 1–2 per hour at nine months to a year. That is the whole market.
- Measured improvement: calibration error down 25–30%.

Caveat: no code is released and the data is paid, so you implement the anchor price and the augmentation yourself. The sample predates the current liquidity regime — re-measure the trade counts before believing them. Pair it with Magadini, Sinclair & Chepal on real-time crypto SVI calibration, which gives you the architecture and problem framing but describes its calibration process as "proprietary". The Kaiko paper gives the maths the Sinclair paper withholds; together they are a working method.

### Two corrections to make before you fit anything

1. **The inverse-contract numéraire.** BTC options on Deribit are coin-margined: premium paid in BTC, settled in BTC. In the BTC numéraire this is not a vanilla — it is a self-referencing payoff, and the implied vol quoted is the USD-numéraire vol of a payoff you never receive. Build a surface from Deribit marks and price a stablecoin-settled structure off it and you are wrong by the correlation between underlying and settlement asset, which here is one. Deribit lists linear options too, and the two families carry different vols for the same strike.
2. **Use the forward, not the index.** Deribit publishes its exact convention, defining the rate as $R = \ln(F/X)/T$ and stating that "the implied volatility displayed uses the forward, rather than the index, as the underlying price." Get this wrong and your implied vols will disagree with the screen by a spread-sized amount, and you will spend a week chasing it.

**The weekend effect is real, and it is not a modelling artefact**

Deribit's own documentation settles it: the exchange uses **pure calendar time, with no business-day or weekend weighting whatsoever**. So the weekend dip in quoted implied vol going into Friday is not something you must undo — it is a genuine, tradeable consequence of the exchange charging calendar theta the market knows will not be delivered. **Whether it survives a 25%-of-premium round-trip on short-dated options is precisely the question, and nobody has published the answer with costs.**

## 8.4 DVOL: an excellent instrument and a non-existent market

DVOL's construction differs from VIX in ways that matter. It uses a **depth-weighted price**, not a mid — walking five levels or two BTC of cumulative size. It **discards ITM options and anything under five delta**, explicitly to avoid strikes where tick size swamps vega. It then applies an exponential moving average over the last 240 one-second points.

What it therefore does *not* measure: it is not a pure 30-day variance swap rate — the five-delta truncation systematically **understates** the true variance swap strike, in the same direction as the mid-price bias above. The EMA means it *lags* by construction and is unsuitable for anything faster than a few minutes. And the depth-weighted construction means DVOL moves when *liquidity* changes, not only when volatility does.

**Is it tradeable? Measured answer: no**

Querying every listed instrument returns exactly **one** DVOL future. Live state at measurement: bid 40.0 / ask 41.0 (a full vol point wide), **open interest 8.1 contracts — about $324 notional — and 24-hour volume of $160**, with one trade that day. There is no ETH DVOL future at all. The listing policy confirms this is by design: one monthly contract.

**Treat DVOL as a data series, never as a hedge.** The valuable part is that the *index history is free and multi-year* at arbitrary resolution, verified returning data for March of 2022 through 2026. That is your variance-risk-premium measurement input.

## 8.5 The perp-option-future triangle is closed

Put-call parity for inverse options gives $F = K/(1-(C-P))$. Comparing the parity-implied forward against the listed future's mark across the whole curve:

| Expiry | Strike | $F$ implied | $F$ future | Difference |
|---|---|---|---|---|
| Front daily | 79,000 | 79,194.03 | 79,183.36 | +0.0135% |
| 1 week | 79,000 | 79,277.47 | 79,253.65 | +0.0301% |
| 1 month | 79,000 | 79,436.90 | 79,422.16 | +0.0186% |
| 3 months | 79,000 | 80,142.02 | 80,111.69 | +0.0379% |
| 7 months | 80,000 | 81,321.47 | 81,324.95 | −0.0043% |
| 10 months | 80,000 | 82,346.89 | 82,334.72 | +0.0148% |

**Maximum deviation across the whole curve: 0.038%, or $30 on $79,000.** Against frictions on the same book: one option tick is $8 to $40, the median front-month ATM spread is 6.67% of premium, and the fee is $23.82 per BTC per leg. **A parity arbitrage needs four legs. The mispricing is an order of magnitude smaller than the cost of one.**

Not exploitable at retail size, and the reason is structural rather than cyclical: three markets on one venue, one margin engine, one index, arbitraged continuously by professionals. **What *is* usable is the inverse** — the parity-implied forward is a free, accurate, per-expiry forward that you should use as the underlying in your implied-vol calculations, exactly as Deribit itself does. That materially changes your fitted skew.

The futures term structure is a clean flat carry curve at about **4.8% annualised** from three months to ten, which is exactly why the triangle closes.

An important tension worth flagging. A published *Journal of Financial Markets* paper on crypto derivative arbitrage reports strategies remaining profitable "even under conservative transaction cost scenarios" — more optimistic than this live measurement. Two likely reconciliations: their sample is older and efficiency has improved, and academic "conservative transaction costs" rarely include the 12.5%-of-premium fee cap or the 9%-of-premium tick that bite hardest on the cheap options where deviations look largest. **Read their profitability claims as an upper bound and re-test on current data with the cost model above.** That re-test is itself worthwhile work.

## 8.6 Where Indian residents can actually trade

### Deribit: not restricted, not registered — and those are different things

Deribit's restricted-jurisdictions list was read directly. It names about twenty-five jurisdictions. **India does not appear.** From Deribit's side, an Indian resident is not blocked. Deribit is also not on the FIU-IND enforcement lists — neither the nine offshore providers served notices in December 2023 nor the twenty-five named in the October 2025 action.

**But Deribit is not FIU-IND registered.** That is a live regulatory gap, not a clearance. Note the direction of travel: the October 2025 action shows FIU-IND enumerating offshore venues serving Indian users and issuing takedowns, and states that the obligations are activity-based and not contingent on physical presence in India. Deribit is now Coinbase-owned, and Coinbase itself registered with FIU-IND. **Treat access as revocable, and do not build anything whose viability depends on Deribit remaining reachable from India.**

And a caveat on the registration claim for the domestic venue: **FIU-IND does not publish a machine-readable public register.** Three official PDFs were fetched and rendered — they turn out to be AML guidelines and the registration *process* circular, not a list of registrants. Aggregate counts come from press releases and parliamentary answers. So a venue's stated registration number is a *self-declaration* that could not be independently confirmed against a government register, because no such public register exists. It is credible but unverified, and you should know the difference.

### The measured depth comparison, which reverses round one

Round two measured both books properly — not five levels, which is meaningless across venues with different ticks, but **notional resting within ±2% and ±5% of mid premium**, which is tick-independent.

| Contract | Venue | Mid $ | bid ±2% | ask ±2% | bid ±5% | ask ±5% |
|---|---|---|---|---|---|---|
| ∼1-day ATM call | Deribit | 437 | **0.0** | **0.0** | **0.0** | **0.0** |
| ∼1-day ATM call | Domestic | 488 | 24.1 | 14.9 | 58.6 | 36.7 |
| ∼1-week ATM call | Deribit | 1,826 | 0.0 | 0.0 | 49.4 | 51.6 |
| ∼1-week ATM call | Domestic | 1,838 | 18.8 | 12.2 | 40.3 | 23.4 |
| ∼1-month ATM call | Deribit | 3,593 | 37.8 | 33.5 | 86.3 | 111.8 |
| ∼1-month ATM call | Domestic | 3,595 | 12.7 | 8.5 | 40.1 | 21.1 |

**The zeros are not missing data.** On Deribit's daily expiry, one tick is already more than 2% of premium, so *nothing can rest inside a ±2% band*. The domestic venue's finer tick makes its short-dated book genuinely tighter — measured spreads of 1.75–2.34% on the one-day ATM against Deribit's 10.9–19.1% on the same contracts.

**The honest reversal**

For **short-dated** BTC options the Indian-registered venue is *tighter and more accessible* than Deribit, because Deribit's fixed-BTC tick is punitive on cheap options. **Deribit wins decisively at one month and beyond** — three to five times the resting notional, 130 strikes against 46 — and wins overwhelmingly on tenor and strike range. Round one had this backwards.

The domestic venue's own book measured about **$486m of open interest and $1.10bn of 24-hour turnover, with every single listed strike two-sided** across nine expiries — reflecting enforced market-maker obligations, against Deribit's 41 of 54 two-sided on its front expiry. Its costs are structurally the same notional-based charge with a slightly tighter cap, *plus 18% GST on the fee*. Lot size is far smaller. There is **no India price premium**: measured simultaneously, spot differed by −0.007% and the perp mark by −0.037%. The folklore applies to INR-funded spot venues, not to a USD-margined derivatives book. There is no cross-venue basis trade here.

## 8.7 Data: what is free, and what you must start recording today

- **Deribit's own API is mostly a dead end for history.** Verified negatives, so you do not waste a week: trade history by currency returned **zero** trades for the same date in 2023, 2024, 2025 and 2026; expired instruments returned only the most recent expiry, so you cannot enumerate historical contracts. **The exception is DVOL** — multi-year, free, at arbitrary resolution.
- **The realistic free deep history** is a first-day-of-every-month options-chain snapshot, downloadable with no API key, with a verified schema carrying microsecond timestamps, *both exchange and local timestamps* (so you can measure feed latency), full greeks, both-side implied vols and sizes, and open interest. Coverage starts March 2019. **Caveat: about 4 GB compressed per day**, and the free days are monthly snapshots — fine for surface cross-sections and calibration research, **useless for event studies or path-dependent backtests**.
- **Consequence: if you want continuous Deribit options history, you must either pay or start recording now.** A single call returns the entire chain with mark implied vols and open interest. Stand up a snapshot recorder today; in six months you own something you cannot otherwise get free.

Honest negative on code, after searching by both relevance and recency: **there is no maintained, credible, open-source crypto options analytics stack.** Almost everything currently active in this space is zero-star, days old and AI-generated. The

one solid recommendation is a maintained, MIT-licensed stochastic-volatility pricing library that is the reference implementation behind the published inverse-options paper — but it is pricing analytics only, with no data layer and no execution. **Given you have already implemented SVI and SSVI, you are better served writing your own thin layer than adopting any of these.**

## 8.8 The crypto variance risk premium, and the gap worth filling

The most rigorous current measurement puts the Bitcoin variance risk premium at about **14% annualised** at the one-month horizon against roughly 2% for the S&P 500, with a Sharpe around 0.8. The regime result is the useful part: **the premium is high in the *low*-volatility regime and lower in high-volatility states** — the opposite of equity intuition. And the return decomposition finds large *positive* returns account for 39% of the Bitcoin premium, against under a third of SPX's premium coming from positive returns at all. **The right tail carries the information in crypto**, which fits the positive front-end call skew measured above.

**But there are no transaction costs anywhere in that paper, and the sample ends before the 2025–26 regime.** A 14% variance premium is comfortably inside the round-trip costs computed above for short-dated options. **Do not read 14% as a strategy Sharpe** — it is a statement about the risk-neutral versus physical gap, not about a harvestable return.

**The highest-value unclaimed work in the crypto half**

Every 2026 claim about the crypto VRP traces to vendor blogs or exchange-affiliated commentary with no published methodology. DVOL history is free back to at least 2022, and minute-wise BTC spot is trivially obtainable. **Compute DVOL$^2$ against subsequent 30-day realised variance yourself, split by volatility regime.** It is a two-day project, it is genuinely original for 2026, and it settles the question with your own measurement rather than an assertion.

# 9. Regime detection and your senior's stack, stress-tested

Your colleague built these. This section is about which survive contact with live trading. The default posture is scepticism, because much of this is academically popular and practically fragile — and where that is true it is said directly.

## 9.1 Hidden Markov models: the weakest link, and the reason is operational

**The bug that accounts for most impressive HMM backtests**

A Gaussian-emission HMM fitted by Baum–Welch gives you *smoothed* state probabilities $P(s_t \mid y_1 \dots y_T)$. Every published equity curve using smoothed states **uses the future to label the present**. The only legitimate object is the *filtered* probability $P(s_t \mid y_1 \dots y_t)$, and filtered probabilities are far noisier and lag the turn by roughly the persistence horizon implied by the transition matrix.

**If your colleague's HMM looks good, the first thing to check is whether it called a Viterbi decode over the full sample rather than running an expanding-window filter.** The demonstration takes about thirty lines: plot filtered and smoothed side by side, plot the equity curve under each, show the gap. That is more persuasive than any paper.

**Second defect: state-count instability.** Information criteria on financial returns are close to flat between two, three and four states, and the argument minimum moves as you extend the sample. A four-state model typically resolves into something interpretable on one window and into a degenerate near-empty state on the next. **A state that exists in one training window and vanishes in the next is not a regime; it is a fitting artefact.** Test it directly: refit on rolling five-year windows and check label correspondence via the Hungarian algorithm on state means and covariances. If labels permute or a state collapses, the model is not deployable.

**Third: Gaussian HMMs reproduce fat tails and zero return autocorrelation but fail to reproduce volatility clustering** — the one property you actually need for options. Flagged as unverified: stated in the jump-model literature and not independently confirmed. It is cheap to check with a simulation and worth doing, because if true it means an HMM regime label is a poor conditioning variable for a vol-selling book precisely because it does not capture the autocorrelation of absolute returns.

### What to use instead

Statistical jump models, with a scikit-learn-style API and discrete, continuous and *sparse* variants. The jump penalty is an *explicit* regulariser on state persistence — you are tuning the thing HMMs leave implicit and unstable in the transition matrix. The sparse variant is the one that matters: it tells you which candidate features actually carry regime information, which kills a lot of feature-fishing.

Real caveat: no evidence any of the authors ran this with client money. "Jump model beats HMM beats buy-and-hold" is an in-sample-of-the-research-programme result, and the jump penalty is a free parameter that does exactly what state count did in the HMM — it moves the overfitting one level up. Tune it inside a purged cross-validation, not on the full sample.

### Who has run it with money

**Honest answer: nobody identifiable**

No named fund could be verified as attributing P&L to an HMM regime filter. The claim that large systematic managers "use regime models" is folklore that could not be sourced, and you should not repeat it.

**Where it stops working is the moment you want it.** Regime detection is a change-point problem with a detection delay bounded below by the signal-to-noise of the feature. In a fast volatility event the label flips *after* the gap move. For a short-gamma book, a regime filter that flips after the gap is worse than useless — it makes you flatten into the bottom.

**Use regime labels only for slow decisions** — position sizing, hedge budget — **never as a gate on entering or exiting short premium.**

## 9.2 Spillover and connectedness: excellent instrument, close to worthless as a signal

What it actually is: a rolling-window generalised forecast-error variance decomposition of a reduced-form VAR, renormalised so rows sum to 100. It is a *measurement* of realised co-movement transmission. It is not a forecast, and the originators frame it as measurement.

The implementations are genuinely good and ship replication data for a battery of published papers, so the field passes tests (1) and (5). It passes neither (2) nor (3): **there is no cost model anywhere in this literature because there is no trade in it.**

### Four failure modes, stated bluntly

1. **The rolling-window artefact.** The index jumps when a crisis *enters* the window and drops when it *exits*, $W$ days later, with no market event at all. Practitioners repeatedly mistake the exit-drop for "spillover normalising". Every such time series has this sawtooth baked in. Time-varying-parameter VAR exists specifically to kill it and is the version to use if you use any.
2. **Generalised FEVD does not identify anything.** It sidesteps ordering dependence, but the price is that directional spillovers are not structural — they are variance shares under a non-orthogonal shock, and rows only sum to 100 after an ad-hoc renormalisation. **You cannot read "A causes B" off the matrix.** Papers do this constantly. Do not.
3. **It is contemporaneous with volatility, so it adds nothing to a vol signal.** Total connectedness rises in crises — so does realised volatility, correlation and VIX.
4. **Out-of-sample instability.** Horizon, lag order and window length are three free parameters and the index level is sensitive to all three. The *shape* over time is far more robust than the *level*, so never threshold on the level.

**The one test that decides whether this survives, and it costs an afternoon**

Before building anything on a connectedness matrix, **regress your connectedness index on cross-sectional realised correlation and average realised volatility.** If the residual is small — and it usually is — the matrix is a repackaged volatility signal with twenty times the maintenance cost. Run this first, and tell your colleague the result either way.

**The one use worth defending** is as a cross-asset diversification *diagnostic*, not a signal. Net directional connectedness tells you which of your positions are net *receivers* of shocks, and therefore where your apparent diversification is fake. For a book short NIFTY vol and long crypto vol, "how much of BTC's forecast error variance is explained by equity and vol shocks in a stress window" is a legitimate, decision-relevant number. Run it monthly, look at the matrix with your eyes, and do not automate a trade off it.

**And note a specific trap for your case:** NIFTY, BTC and SPX do not share a session. A VAR estimated on misaligned bars creates spurious directional spillover *from the market that closes later*. That is a mechanical artefact, not a finding.

## 9.3 Clustering for parameter selection: it does not do what it claims

**The honest framing**

Picking a "robust plateau" in parameter space via clustering is a *smoothing operator* on the backtest-Sharpe surface. Smoothing reduces the variance of the selected parameter's out-of-sample performance and slightly reduces selection bias versus taking the argmax. **It does not address the fact that you ran $N$ trials.** If you tried 500 combinations and then clustered them, you still tried 500 combinations, and your best cluster centroid still needs deflating by the same multiple-testing correction.

**Clustering that just relabels overfitting is exactly the right suspicion.**

**The recursion is the fatal objection.** A density-based clusterer has its own hyperparameters — so using it to pick your strategy's hyperparameters means you now have hyperparameters selecting hyperparameters, tuned on the same data. People then Bayesian-optimise *those*, adding a third layer.

Each layer is another set of trials that nobody counts.

A nearest-neighbour approach for this purpose is simply local averaging of the Sharpe surface. If that is what you want, say so and use a Gaussian kernel smoother — one line, one bandwidth parameter, and honest about being a smoother.

A targeted search found *no* credible finance practitioner source applying density-based clustering to strategy-parameter selection. What comes back is microscopy, astronomy and the documentation.

**What to do instead:** drop the clustering. Replace it with (a) sensitivity of Sharpe to each parameter one at a time, plotted; (b) a requirement that the strategy be profitable across a *contiguous region*, not at a cluster centroid; and (c) **count every trial you ran, including the ones you abandoned, and deflate**. Item (c) is the one everyone skips and the one that matters.

One caveat that cuts against the standard advice for *your* case: purged cross-validation matters most when labels overlap. **For a minute-wise intraday NIFTY options strategy that opens and closes within the session, label overlap is near zero and purging buys you almost nothing.** Your real leakage risks are different — expiry-day survivorship in the chain, strike-selection using the day's realised ATM, and quoting off a stale mid.

## 9.4 Delta-neutral construction: this one survives

It is the only item in your colleague's stack where the cadence question is genuinely solved with closed-form answers you can implement this week. That material is Section 5 and is not repeated here. The single most important point to carry back:

**The correction your colleague may be missing**

The optimal response to transaction costs is **not** "hedge less often on a timer". It is a **no-trade band around delta whose width scales as the cube root** of (cost × gamma × spot / risk aversion). Time-based rebalancing is dominated by band-based rebalancing at essentially every cost level, and the gap widens with gamma. With minute NIFTY data you can measure this directly.

**Where it stops working:** the band solutions all assume a diffusion. In a gap — budget day, RBI policy, a crypto liquidation cascade — the framework is silent and the gamma cost is realised instantly regardless of cadence. And on expiry day the asymptotic assumptions break entirely (Section 5).

**The crypto-specific gap worth knowing:** Deribit is 24/7 with no weekend gap, so the diffusion assumption behind the bands holds *better* than in NIFTY — but perpetual funding introduces a carry term in the hedge leg that none of the classical band literature models, and that term can dominate the transaction-cost term. That is unfilled, and it is a genuinely publishable piece of work.

## 9.5 Trend following and options are the same trade

The best-founded item in this section, and the only one with a named person managing real money who published the analysis free.

The identity is operational, not decorative. Replicated straddle P&L equals the payoff range minus $\frac{1}{2}\sum \Gamma_{s-1}(S_s - S_{s-1})^2$, and **that quadratic-variation term is named as the slippage cost of replication**. Theta in the option is identified as the *same quantity* as whipsaw plus transaction cost in the trend program. That gives you a direct arithmetic test:

**A runnable decision rule**

Compute $\frac{1}{2}\sum \Gamma(\Delta S)^2$ for your trend rule and compare it to the actual traded straddle premium. **Whichever is cheaper is how you should express the trade.** You can run this on minute NIFTY data.

The rest of the mapping: the moving-average crossover threshold is the straddle strike; position-sizing ramp is delta; ramp speed is gamma. **Strangles correspond to entry thresholds** — spreading the strikes is exactly "require confirmation before entering", and the stated cost is slower convexity accrual, "the cost of patience". **Stop-losses correspond to knock-out options**: a trend program with stops is *short a tail put below the stop level* and exposed to jump risk there. The prescription, if the option is cheap enough, is to replace the stop with the option outright.

The caveat that matters: **there is no backtest in that paper.** No cost-modelled equity curve, no parameter set, no data

appendix — it is an analytic framework plus illustrative exhibits. It also has an obvious commercial interest: a tail-risk manager arguing that trend followers should buy options is arguing for its own product. The framework is correct and useful regardless, but do not treat it as evidence that the overlay pays after costs. **You have to build that test, and you can**, because the whole thing reduces to comparing $\frac{1}{2}\sum \Gamma(\Delta S)^2$ against traded straddle prices. To their credit they volunteer the negative: over the most recent two decades their 50/50 allocation did *not* keep up with 100% equity on CAGR and won only on risk-adjusted terms.

### The finding that invalidates most options-overlay backtests

**Rebalance timing luck**

A self-financing three-month put-spread collar was replicated in implementations differing *only in rebalance date*. The result: **annualised tracking error in excess of 400 bp between implementations.** For equity factor indices the luck term exceeded 100 bp a year.
**Any options overlay you backtest on a fixed roll schedule has a several-hundred-basis-point random component from the roll date alone.** The fix is overlapping tranches — roll a fraction of the book each period. If your backtest platform rolls on a single fixed date, its results are not reproducible in the sense that matters.

## 9.6 The empty `regime` column in your platform

Not an oversight to fix by picking a model. It is a schema stub, and what it should hold depends on a decision nobody has made. In production systems this field is normally one of four things:

1. **A low-cardinality categorical label per bar** from a small enumerated set. Almost always *not* from an HMM — it is a deterministic rule over slow observables: trailing realised volatility bucketed by its own trailing percentile, term-structure slope sign, a drawdown-from-high flag. Deterministic rules are used because they are auditable, have no refit, and cannot silently leak.
2. **A continuous score plus a threshold**, stored as the score, bucketed at query time. Preferred by teams burned by state-count instability.
3. **An as-of-bar filtered probability vector** with an explicit model version and fit-as-of column. The presence of a fit-as-of column is the tell that whoever built it understood look-ahead.
4. **A post-hoc analyst label** used for slicing P&L reports and explicitly forbidden from entering signals.

**The distinction to enforce before anyone populates it:** (1) to (3) are point-in-time and legal as inputs; (4) is retrospective and is an attribution field only. If it is going to be filled, it needs a sibling boolean or the two uses will get mixed within a quarter — and a smoothed retrospective label leaking into a signal is exactly the bug at the top of this section.

**The useful thing to say in the meeting**

**An empty regime column is a better state than a wrongly populated one.** Ask what decision the field is supposed to change. If the answer is “position size” or “hedge budget”, a deterministic volatility-percentile rule is sufficient and costs a day. If the answer is “whether to enter short premium”, **the field should stay empty**, because detection lag makes it actively harmful there.

# 10. People, books, and where to actually read

## 10.1 The seed video, audited

You sent one video as the exemplar. It is worth stating plainly what the audit found, because the pattern generalises and recognising it will save you far more time than any single recommendation here.

**The exemplar fails the filter on five of six counts**

- Primary call to action is a paid coaching programme at roughly $3,000 to $5,000, **including trade alerts and algorithmic signals**. Course plus signals is the exact shape the filter excludes.
- Brokerage-sponsored with an affiliate link, and carrying the platform's own paid-promotion badge.
- **The "verified performance" link does not survive a click.** The linked portfolio shows a name, a city, a Verified badge — and then "Trades are hidden for this user." No equity curve, no trades, no drawdown. The verification is a badge with the substance switched off.
- **The trade evidence is about 23 seconds.** By the video's own chapter list, "Real Trade Examples" runs 10:32 to 10:55. The other nine minutes are whiteboard mechanics.
- Every title in the channel is a dollar amount plus a win rate — 89%, 91%, 92%, 96% — advertised without loss-size context, which is precisely the metric that flatters short premium and hides the tail.
- The channel is fourteen months old and already sells a multi-thousand-dollar programme.

**To be fair to it: double calendars are a completely legitimate structure**, and the mechanics section may well be sound. The problem is the register and the unfalsifiable record, not the instrument.

**The generalisable tell**

**Any claimed track record whose per-trade fills you cannot inspect is not a track record.** Several better-credentialled US names, and most of the Indian option-selling ecosystem, are the same object at different production budgets.

### What to read instead, on the same subject matter

| Question | Honest source |
|---|---|
| Does premium selling have an edge at all, and how do I size it? | Sinclair's position-sizing work — the only treatment that starts from "here is the cost of being wrong about the fourth moment" rather than from a win rate. |
| Double calendars and diagonals | **There is no good dedicated free source, and that is informative.** The correct framing is *forward variance and term-structure slope*, which you already have. A double calendar is a bet on forward vol between two expiries plus a short skew position — price it as such and the folklore evaporates. Nobody teaching calendars as a "range play" is pricing the forward vol. *And on NIFTY, per Section 3.3, there is barely a second expiry to trade.* |
| Delta-neutral short vol: the failure modes | Kris Sidial on liquidity gaps and why the hedge does not fill; Benn Eifert on why the structural short is a levered carry trade; and the QVR wind-down itself. |
| The actual net cost of doing this in India | SEBI's studies — roughly ₹25,000 crore a year of transaction costs is the number every Indian income-strategy backtest omits. |

## 10.2 Practitioners, gated on a verifiable CV

Verification standard: **V** means confirmed against at least one source independent of the person's own marketing; **S** means self-reported or repeated only by promotion-adjacent outlets.

### The highest-value names

**Euan Sinclair** — *the single highest-value name here for you.* PhD theoretical physics; roughly 25 years professional options trading; risk management and strategy design at a proprietary options firm. **V**. His free SSRN papers are directly implementable: the risk-reversal premium, the skewness and kurtosis of European option returns and the implications for trade sizing, volatility estimation via first exit times. **The trade-sizing paper is the one to read first** — it connects the third and fourth moments of an option's own return distribution to position sizing, which is exactly the gap between your surface work and an actual position. Passes tests (2), (3), (4) doubly, and (5). *Caveat: US-centric, and some of the specific premia have decayed. Transfer the method, not the parameters.*

**Timothy Klassen** — see Section 5. Goldman under Derman, designed Cboe's 2003 VIX, built Getco's options analytics. **V**. The richest free corpus in this document for spot-vol dynamics and P&L attribution. *Caveat: vendor, selling a library, and his SVI criticisms are made by an interested party — benchmark them against your own NIFTY fits.*

**Kris Abdelmessih (Moontower)** — SIG trainee, then market maker across several option pits, then a relative-value commodities vol book. **V**. The best free writing anywhere on thinking in payoff space, and repeatedly explicit about which retail vol ideas are simply the bid-ask spread in disguise. *He writes for intuition, not rigour — almost nothing there is a finished testable specification, and converting his intuitions into hypotheses is your job. Skip roughly 40% of the Substack, which is personal-finance and lifestyle.*

**Kris Sidial (Ambrus)** — co-CIO of a live volatility-arbitrage and tail-risk fund; previously an exotic derivatives desk at a bank and proprietary trading. **V**, corroborated across several independent outlets. Unusually explicit about *how a tail hedge actually fills* — liquidity, cost of carry on convexity, and the fragility of options-market plumbing. Passes test (4) hard: he names where his own trade fails, which is that a slow grind-down bleeds him out and he needs discontinuity. No paid product for retail.

**Cem Karsan** — mostly verified, with one claim that does not check out. Founded an SPX volatility-arbitrage market-making firm; his current firm is a registered commodity trading advisor, which is a regulator record rather than a bio. **V**. *But the widely circulated "32 years of exchange trading experience" is arithmetically impossible against a 1999 undergraduate degree — it appears to be firm-level language that migrated into his personal bio via speaker directories. His real book is roughly 2000 to present, which is still about 25 years. Marketing drift, not fraud, but he has never corrected it.* **Good for** the mechanical, flow-based half of index vol. **Ignore** the macro and political narrative layer — it is unfalsifiable and unrelated to the part of his work that has edge.

**The severe caveat on Karsan, which is load-bearing**

**He has never published the computation.** In none of the free long-form does he specify how gamma or vanna exposure is estimated from open interest, what the dealer-sign assumption is, or how it is validated. **The framework as he presents it is unfalsifiable as stated**, and the commonly cited underlying methodology is anonymous and attached to a paid data product — an auto-reject on two counts under your own filter.

**Read Barbon & Buraschi's *Gamma Fragility* first** — free, named, peer-reviewed, and it finds intraday momentum under negative aggregate dealer gamma and reversal under positive, *conditional on illiquidity*. That conditioning is the failure mode Karsan never states. Then read Karsan as colour. With NIFTY open-interest data and minute bars this is directly replicable on an Indian index and, as far as can be determined, unpublished there.

**Benn Eifert (QVR)** — PhD economics; head of quantitative research and derivatives trader on a bank proprietary desk; co-founder of a real vol-arbitrage manager. **V**. Years of free, technically precise threads on structured product flows, dispersion, how bank vol supply distorts surfaces, and a sustained public demolition of retail "options income" strategies.

**The most instructive single fact in this section**

**QVR's multi-strategy fund lost roughly 30% between January and April 2026 and is winding down**, from a peak around $1.6bn, with the management business up for sale. This does not invalidate his analysis. It is the empirical footnote: **the most technically credentialled, most cost-aware, most failure-mode-literate vol manager in the retail-visible world still blew up.** Read the back catalogue *and* read the blow-up. Do not treat anyone's process as proof of survivability — including your own.

**Harel Jacobson (volquant)** — roughly 15 years in FX volatility trading; all free, no paywall, no funnel. **The closest thing to a free written market-maker course.** His FX volatility smile series matters to you even though you trade index: **Part II explains *why* the smile has the shape it has as a consequence of dealers hedging exotic books** — risk reversals to hedge vanna from barriers, butterflies to hedge volga from digitals. That is a supply-side explanation of skew and convexity that no equity-options text gives you. One piece includes runnable code, so he passes test (1). *Caveat: FX conventions do not transfer without translation, and he says so.*

### Verified and rejected, with reasons

- **Imran Lakha** — background is the cleanest of any candidate (**V**: Citigroup head of European index options, then head of a bank's European options desk, then macro PM). **Auto-rejected on outlet**: the primary funnel is a paid membership plus a course. His free weekly vol commentary remains usable as *market colour* — how a sell-side index-vol book thinks about skew and implied correlation — never as a source of a testable claim, and the free tier is deliberately incomplete.
- **A well-known calendar-and-butterfly educator** — CV is **S** only: every restatement traces to bios he supplied, and no independent primary record was found. Auto-rejected on the paid mentoring funnel. His pre-course-era webinars are pedagogically decent and contain no reproducible testing. He is the closest US analogue to the video you sent — better credentialled, same commercial structure.
- **A prominent proprietary firm's "options desk" — reject, and the implied pedigree is weaker than presented.** What actually checks out (**V**) materially contradicts the marketing impression: the lead is a career property-and-casualty insurance executive who left insurance in 2006 and has traded options since — not an ex-market-maker or ex-desk trader. The firm is real; the options *desk* is substantially an education product. Skip entirely.
- **A large brokerage's media arm** — the founder's CV is genuine and large (**V**: about 20 years as a floor market maker, founded two brokerages). **Auto-rejected twice**: it is the media arm of a broker, so order flow is literally the business model, and their "studies" are the marketing. Documented failures to reproduce their backtests exist. Use it for vocabulary and the *mechanics* of managing a short-premium book — rolling, assignment, margin — and then test every quantitative claim yourself. **Do not treat a single one of their numbers as an input.**

## 10.3 Indian practitioners: the honest answer

**Almost nobody survives, and the bar was not lowered to manufacture entries**

Checked and rejected: the major broker-integrated platforms (order flow is the business model, and one widely repeated "ex-Goldman" claim does not check out); the best free Indian options primer, which is broker-affiliated and also below your level; the largest quant-education blog, whose parent is a genuine high-frequency firm with verifiable people but which exists to funnel into a paid programme; and the closed-box backtesters, several of which *do* let you set slippage and brokerage — more than most US retail content offers — but whose fill logic you cannot inspect, so nothing produced there is reproducible.

**And the structural problem: the serious Indian quant firms are real, with verifiable people, and they publish nothing.** The competent Indian options community is entirely inside proprietary firms and does not write.

**Two Indian sources do pass, and they are not people:**

1. **SEBI's studies on individual traders in equity derivatives.** The only Indian document that states the true all-in cost drag. It tells you precisely who is on the other side of your fills and what the toll booth costs. *Caveat: regulator aggregate data with a policy agenda; it tells you the population loses, not why any individual strategy fails.*
2. **Academic work on Indian market microstructure** by the people who built the institutions — foundational quantitative work on Indian microstructure and index construction, published free, plus first-person writing on Indian derivatives structure. *Caveat: policy and microstructure economics, not a trading strategy, and much of the derivatives-specific work is from the late 1990s and 2000s. Read it for how NIFTY liquidity and impact cost are actually structured.*

**The blunt conclusion**

There is no Indian equivalent of Sinclair or Abdelmessih. Given that you will hold minute-wise NIFTY and Sensex data and can model costs against SEBI's published aggregates, **the honest recommendation is to import technique from the US and crypto names above and become the missing Indian source yourself**, rather than keep searching for one.

**The same applies to video.** All Indian long-form options content is broker-affiliated education, course funnels, or "I made ₹X" content. For Indian-market-specific craft there is *no free practitioner video layer, and you should not go looking for one.* Your Indian inputs should be exchange and regulator primary documents plus your own data; the technique layer should come from elsewhere, translated.

## 10.4 The book canon, ranked for your position

Ranking premise: you have the surface maths. What you lack is (i) how a book behaves over time, (ii) what the costs actually are, (iii) how to tell whether a model is broken. Ranked by marginal value *given that*, not by general quality.

### Access reality first

**Do this before spending anything — it is a two-minute email**

The national journals consortium your library leads with is a **journals** deal and does not cover monographs. Indian institutional subscriptions typically include a publisher's *journals*, not its *online books* — and Sinclair, Taleb, Gatheral, López de Prado and Rebonato are all with the same publisher, so the likely outcome is that **none of them is reachable and you buy them**. Springer/Palgrave e-books are commonly held, which makes one title on this list most likely free to you. McGraw-Hill Professional is effectively never in an Indian academic package.

**Ask the library for the e-books A–Z list, and specifically whether the publisher's *Online Books* and Cambridge Core *Books* — not just journals — are in the subscription.** That single question resolves roughly ₹40,000 of purchasing decisions. Flagged: the library site is login-gated, so all access claims here are inference from publisher and consortium patterns, not verification.

### Tier 1 — in this order

**1. Sinclair, *Volatility Trading* (2nd ed.).** The only book that treats an options trade as a *statistical bet with a cost structure* rather than a payoff diagram. The hedging chapter derives the path-dependence of a delta-hedged option's P&L explicitly — why your realised P&L is not the vol difference times vega but a gamma-weighted path integral. The money-management chapter is where most quant-trained people are weakest. **Chapters that matter:** volatility measurement, hedging, money management, trade evaluation. **Skippable:** the Black-Scholes derivation and the options-basics chapter. **Dated:** pre-0DTE, US-centric, and its GARCH-family forecasting chapter is the weakest part. **No free equivalent exists.** This is the book with no substitute.

**2. Bergomi, *Stochastic Volatility Modeling* — and chapters 1 and 2 are free on the author's own site.** The correct answer to "what happens to my book when spot moves". Chapter 1 is titled around *How (in)effective is delta hedging?* and derives the P&L of a delta-hedged short option from break-even

accounting, then shows empirically how much hedging error survives. That is the attribution content, free, from a bank's head of quantitative research. **Chapters that matter:** 1 and 2 (free), then the smile-of-stochastic-vol chapter where the skew-stickiness ratio is derived, and the static-versus-dynamic and what-causes-equity-smiles chapters. **Free equivalent:** the two free chapters plus his *Smile Dynamics* papers cover perhaps 40% of the practical value at zero cost — **read those before deciding to buy. Caveat:** dense idiosyncratic notation and no code. You will be re-deriving.

**3. Taleb, *Dynamic Hedging*.** The *bookkeeping* of an options book — how a market maker organises risk across hundreds of positions in one name. The only book-length treatment of **vega bucketing** by someone who did it on a desk, and the only serious treatment of what happens *between* the greeks: pin risk, wing risk, early-exercise pathologies. **Dated and in places wrong:** it predates SVI, SABR, everything you have implemented, and has no coherent spot-vol dynamics — precisely the gap Bergomi and Klassen fill. Its exotics pricing is obsolete. **Read it for the taxonomy of risk, not for models.** Badly organised and badly edited; budget for frustration. Nothing has replaced it.

**4. Bouchaud et al., *Trades, Quotes and Prices*.** The cost side, quantitatively: square-root impact, order-flow long memory, the propagator model, why your backtest's fill assumptions are the dominant error term. **Real free equivalent:** Bouchaud, Farmer & Lillo, *How Markets Slowly Digest Changes in Supply and Demand* (arXiv:0809.0822), about 100 pages covering a large fraction of the relevant parts. **Read that before buying. Caveat:** options microstructure is essentially absent, which matters because NIFTY option books behave differently from the underlying.

### Tier 2 — conditionally

**Sinclair, *Positional Option Trading*.** The value is the negative result: most named option "strategies" carry no edge, and the edge where it exists is in sizing, selection and exit. Keep the bet-sizing and process chapters; skip the strategy catalogue. Substantial overlap with *Volatility Trading* — buy second, not first.

**Gatheral, *The Volatility Surface*.** You have absorbed the parameterisation content already. What remains valuable is the skew-dynamics chapter and the variance-swap chapter. **His free Baruch lecture notes overlap the book heavily — use the notes, skip the book.**

**Natenberg, *Option Volatility and Pricing*.** For you, roughly 70% is below your level. The residue that is not: how a spread's greeks migrate as spot and time move, and the market-maker chapters. **Fails test (2) — no cost model anywhere. Recommendation: skip the purchase.**

**A short Palgrave monograph on the greeks and hedging** whose publisher description names exactly the gaps above — day conventions, interest rates, sticky deltas. **The title on this list most likely to be free to you.** Caveat: thin; it will not replace Bergomi.

**López de Prado, *Advances in Financial Machine Learning*.** Include it *only* for four things: triple-barrier labelling, meta-labelling, purged/embargoed cross-validation, and the deflated Sharpe ratio. Everything else is contested. **Nearly all the substance is in his free SSRN papers.** Honest negative: a targeted search for a substantive published critique of this book's reproducibility found none from a credible named practitioner — the criticism circulates as forum folklore, not as a citable document. Treat its methods as *unreplicated* rather than either validated or refuted.

### Tier 3 — do not buy

**Rebonato, *Volatility and Correlation*** — the majority is interest-rate volatility, irrelevant to NIFTY index options, and the relevant parts are superseded by Bergomi. **Sinclair, *Option Trading*** — largely superseded by his other two. **A 2024 retail-oriented options title** — given your level, skip; credible review describes it as oscillating between explaining what a long call is and computing rolling volatility z-scores.

**Books that earn a slot and were not on the original list:** Derman & Miller, *The Volatility Smile* (though Derman's Columbia lecture notes are free and cover most of it); Larry Harris, *Trading and Exchanges*, still the canonical market-structure text and cheap; Lehalle & Laruelle, *Market Microstructure in Practice*, the execution-cost companion to Bouchaud.

# 11. Income and honest base rates

Factual only. This section reports what is documented — capital levels, cost structures, published loss rates, and which skills monetise where. It contains no recommendations, no allocations and no trade suggestions.

## 11.1 Two facts that reorder the question before any maths

1. **US options are closed to you as a venue** (Section 3.1). Anything you read about US retail options economics is technique transfer, and the constraint is legal rather than stylistic.
2. **The crypto tax regime is anti-selected against exactly the strategies you are training to build** (Section 3.2), and the dominant crypto options venue is not FIU-registered.

So of the candidate strategy classes, **two are venue-constrained before you reach a line of maths**, and the realistic choice set is NSE index derivatives versus Indian-domiciled crypto venues.

## 11.2 Capital versus viability, with numbers

### Indian index options: the floor is the flat fee, then the margin

The ad-valorem charges are not the problem at small size — a one-lot round trip on a short option costs roughly 1.3% of a good gross P&L. **The flat per-order brokerage is the problem**, because it is per *order*, not per lot, and so dilutes about tenfold going from one lot to ten.

A four-leg iron condor is eight orders. At one lot, brokerage plus GST alone is roughly 7% of maximum profit — and if you exit at 30% of maximum, which is the standard management rule, **costs are about 23% of the realised target**.

**The two capital floors**

**Defined-risk multi-leg selling** becomes fee-viable at roughly ten lots, which is about ₹1.5–2 lakh of margin. **Undefined-risk selling** has a much higher floor driven by margin, not fees: from the computed table in Section 7.4, **₹1.52–1.84 lakh per lot for a short strangle and ₹2.62 lakh for a 0DTE short straddle**. Add the requirement that at least half of margin be cash or cash-equivalent, so a ₹20 lakh "account" is not ₹20 lakh of deployable margin.

**Scaling arithmetic — division, not a return assumption.** At a net 1% a month on deployed margin you need roughly ₹1 crore of margin to gross ₹1 lakh a month; at 2% a month, about ₹50 lakh. Then F&O profit is business income at slab. **There is no configuration of this arithmetic in which a sub-₹25 lakh account produces income competitive with a finance salary inside two years.**

**And the regulator is an active adversary of the small-account version.** The November 2024 package — contract size raised to ₹15 lakh-plus, weeklies cut to one index per exchange, STT raised, then raised again in April 2026 — was designed to raise the cost floor for exactly this activity.

### Crypto

The lowest capital floor mechanically — no lot size, fractional sizing — with the floor set by fees, transfer costs and margin buffer. **Then the tax code eats it** (Section 3.2): for an Indian resident the after-tax carry can be negative while the gross carry is positive. That is the single most under-appreciated fact in this section.

For crypto *options* specifically, serious-participant treatment on the deep venue starts around six figures of account equity and market-maker terms require a negotiated agreement — and it is not the registered venue anyway.

## 11.3 The base rates

### SEBI — and note the numbers you had are superseded

| | FY24 | FY25 | FY26 |
|---|---|---|---|
| Net losses (₹crore) | 74,812 | **1,12,000 (revised)** | **91,685** |
| Loss-maker share | 91.1% | **90.9%** | **87.7%** |
| Unique individual traders (lakh) | 86.3 | 96.0 | **87.5** |

The August 2026 study revised FY25 *upward* from ₹1.06 to ₹1.12 lakh crore — so the figure in circulation is stale. FY26 shows the first year-over-year decline in active traders since FY16, down 18%, with new entries down 40%. **But average loss per person rose 2%**, and 23% of traders account for about 90% of total losses.

Options accounted for **92% of aggregate individual losses**, and 93% of individuals traded *only* options. **99% of foreign-investor and proprietary profits came from algorithmic entities.**

**A claim from round one that must be retracted**

Round one inferred from these studies that "a large share is naive short vol." **The studies cannot support that and do not attempt to.** A full-text search of the current study for direction-related terms returns six hits in 91 pages, none about strategy composition. The methodology table lists the complete categorisation — product category, age, income, gender, city tier, turnover, portfolio size, frequency, experience. **There is no direction field. No long/short. No buy/write.** The sign of the position is never in the dataset, and SEBI states its own limit: "the analysis is descriptive and does not establish cause-and-effect."

### And SEBI has now published the document that settles it

A companion *behaviour* study classifies traders by strategy using clearing-corporation settled-value, traded-value and margin data, on a random sample of 5,050 individuals. FY26:

| Category | Share of traders | Loss-makers | Avg. loss per loss-maker |
|---|---|---|---|
| Only options buyers | **93%** | 90% | ₹1.3 lakh |
| Majorly options buyers | 4.2% | 75% | ₹4.6 lakh |
| **Majorly options sellers** | **2.1%** | **44%** | **₹51.7 lakh** |
| Majorly futures traders | 0.9% | 61% | ₹1.7 lakh |

**The retail base is 97% option *buyers*, not sellers**

The sellers are the smallest group and have the *lowest* loss rate — and lose eleven times more when they lose. The category still net-lost money in FY26. SEBI's own framing: "relatively high win rates may mask a tail-risk profile."

**This is the single most important number in this document for your risk management**, and it points the same way as everything in Section 5.3: the danger in short premium is not the mean, it is the skew.

Honest caveat, which SEBI states itself three times: the seller cell is **105 traders in FY26**, of whom 46 were loss-making. The ₹51.7 lakh average is a 46-observation mean of a fat-tailed variable. **Treat the direction as solid and the magnitude as indicative.**

### The two studies that do what SEBI's cannot

**Brazil, day traders in index futures.** Regulator records on 19,646 individuals who began day trading, followed for years. **97% of those who persisted more than 300 days lost money**; 1.1% earned more than the minimum wage; 0.5% more than a bank teller's starting salary, and those with high risk. **No evidence of learning by doing.** *What it does not show:* a single instrument, a single style, retail with no stated prior training. It says nothing about multi-day holding periods or people entering with

institutional-grade quantitative preparation. The "no learning" result is the part that generalises least comfortably and is also the part most worth sitting with — it is the direct empirical rebuttal to "I will be profitable once I have done my two years."

**Taiwan, full population, 1992–2006 — the only study here with a proper persistence design. Less than 1% of day traders predictably and reliably earn positive abnormal returns net of fees.** Predictably profitable traders were under 3% and generated only about 10% of volume. Sorting on one year's returns, the top 500 went on to earn +37.9 bp a day after fees the following year; the bottom ranked −28.9 bp.

**The most useful number here for you**

That study says skill is **real, persistent, and detectable** — and concentrated in a sub-1% tail. It also says the persistent winners are *identifiable in advance from track record*. So the honest test is empirical and self-administered: **does your own live, cost-inclusive record show persistence?** Nothing else answers it.

**Proprietary-firm evaluations.** The best available dataset — roughly 300,000 accounts across 100,000 traders at ten firms — gives about **14% passing an evaluation**, about **45% of funded traders ever receiving a payout**, so roughly **7% of everyone who buys a challenge ever sees money back**, and 1–3% still withdrawing beyond six months. Structural caveats matter more than the percentages: most are simulated accounts, the challenge fee is the revenue model (which aligns the firm against your passing), and the model is forex and futures — **it does not test the skill you have.**

**The study you want does not exist**

There is **no published longitudinal study of the population you belong to** — a quantitatively trained person with a full-time job trading own capital systematically. Brazil and Taiwan study retail day traders. Anyone who tells you the base rate for your case is extrapolating.

## 11.4 Solo versus firm, per skill

| Skill | Solo? | The honest reason |
|---|---|---|
| Vol surface modelling | **No, not directly** | It is an *input*, not a strategy. At a firm it is table stakes — every desk has a calibrated surface and nobody is paid for having one. Its solo routes are intra-surface relative value, which needs quoting infrastructure you cannot access, or generating a rich/cheap signal, which collapses back into premium selling. **You have already built the thing; the thing is not the edge.** |
| Regime detection | **Yes** | The only cleanly solo-scalable item — no infrastructure, no latency, no venue relationship. **Blunt counterweight:** it also has the weakest out-of-sample evidence base in all of quant finance (Section 9). |
| Market making | **No** | Requires exchange membership or a negotiated agreement, quoting obligations, colocation, rebate-tier economics and inventory capital. **There is no retail version.** Every "retail market making" claim is wide-spread scalping or marketing. |
| Statistical arbitrage | **Partially, and worse than it looks** | Indian cash shorting is intraday-only and stock lending is thin, so you cannot hold the short leg. Crypto cross-venue is reachable but capital- and infrastructure-hungry, margins have compressed, and the tax regime taxes the gross legs of a spread independently. |

| | | | |
|---|---|---|---|
| Options selling | premium | **Yes — fully** | Which is exactly why it is crowded, and exactly what the SEBI numbers measure. The one item an individual with margin can run at institutionally identical mechanics. **The risk is not mean edge — it is skew.** A single gap can exceed multiple years of carry, and nothing about surface competence protects against it: the exposure is to the jump, not to the surface. |

**The uncomfortable summary, in one line**

**The skill you have invested most in — surface modelling — is the least solo-monetisable. The one that is fully solo-monetisable — premium selling — is the most crowded and the most tail-exposed.**

## 11.5 Quant-adjacent income that is not a software job

Assessed honestly; most of it does not pay on your stated horizon.

**Crowdsourced quant platforms.** Real money in aggregate — tens of millions distributed — but the distribution is brutal: of roughly 14,400 models submitting in a trailing year, only about 2,600 finished net positive, with a *median* positive earner around $1 and a 90th percentile around $235. And you must stake your own capital in a volatile token. **Verdict: a hobby with a lottery tail**, correctly classified alongside the trading base rates rather than alongside income.

**Hosted quant competitions.** Prize pools typically $100k–$120k split among the top handful of several thousand teams. The realistic return is a credential and a recruiter conversation. **Given you are excluding a job, the credential has zero cash value to you.**

**Systematic-strategy licensing — the flagship attempt failed.** The most serious retail alpha-licensing marketplace ever built was wound down: sales disabled, live hosts stopped, licensees refunded. The reason is the one that matters — **there were not enough buyers. Institutions do not license black-box alphas from anonymous individuals.** What remains are performance-fee and subscription-share platforms, which are real and structurally sound; the binding constraint is *attracting third-party capital*, which is a distribution and track-record problem, not a quant problem. **The gate is a two-to-three-year live, audited, cost-inclusive track record before anyone subscribes — which consumes your entire stated horizon before the first rupee. It is not a two-year path; it is a path that *begins* at year two.**

**Paid research writing.** Real but small and brand-gated. The people earning meaningfully almost universally arrived with an institutional brand first. Writing does not bootstrap the brand quickly enough to matter on a two-year clock. No verified income distribution data exists for this; reporting the absence.

**Data products — a genuine gap you are structurally blocked from filling.** Indian minute-level index options data is expensive and poorly served, and there is real demand for cleaned implied-vol surfaces. But redistributing exchange-derived data requires an authorised data-vendor licence, and **the minute-wise data you are about to receive comes through an employer relationship — you cannot resell what you get at work**, and attempting to would be the fastest possible way to end both the job and the path. Closed for reasons that have nothing to do with skill.

## 11.6 The arithmetic nobody states plainly

**Capital, not skill, is the binding constraint at your stage**

A ₹25 lakh account returning a genuinely excellent, sustained **20% net** produces ₹5 lakh gross and roughly **₹3.5 lakh after slab tax** — a fraction of a finance salary, from a strategy performing at a level the Taiwan study places in the sub-1% persistent tail.

To make the income question interesting rather than academic, the account has to be an order of magnitude larger, and **the salary is the only documented mechanism that gets it there inside two years.**

That is division, not advice.

Employer personal-account-dealing policies vary substantially — pre-clearance requirements, minimum holding periods,

restricted instrument lists, and outright bans on instruments the firm covers. Check whatever contract you eventually sign before any of the above is actionable.

# 12. Current chatter — UNVERIFIED

**Read this section differently from the rest**

Nothing here passed the source filter. This is a **routing list** — names, repositories and, most valuably, *complaints* — to be sent to independent verification. Nothing in it is evidence that anything works.

Method disclosure. The dedicated social-sweep tool was invoked and produced unusable output: incomplete first-run setup, no authenticated access to X, rate-limiting on Reddit after six items, and a keyword collision on "options" that surfaced consumer-electronics threads. The fallback used the GitHub and Hacker News APIs, a Reddit archive mirror, and web search. **X and YouTube were not swept at all**, which is a real hole — see the coverage failure below.

## 12.1 The complaints, which are the valuable part

- **Gamma-exposure services are being publicly demolished.** A widely upvoted thread and a companion study report that spot itself beats gamma-exposure and max-pain as a predictor for index 0DTE. One participant claims a year of the full greek-exposure dataset with a null result. The mechanism offered matches Section 4.4 exactly: most services compute exposure from *open interest with no participant tags*, so the dealer sign is guessed.
- **The variance risk premium in index 0DTE is real but "razor thin".** One participant ran tick-level ATM implied vol against per-minute realised vol over 280 trading days and reports roughly **3–4 vol points of premium, strongly regime-dependent, with extended runs where it is absent**. This is the most methodologically serious retail work found in the window, and **the method transfers directly to NIFTY minute data.** Route it to verification.
- **Regime switching "sounds good but disappoints"**, in a thread of that exact title. The sharpest comment: one needs to detect a regime switch *before* it happens, and if you could do that you would be rich. This is the community arriving independently at Section 9's conclusion.
- **Latency is where backtests die, and most tools cannot see it.** The recurring point: the gap between signal time, order time and exchange receipt is longer than people think, and most backtesting tools have no visibility into it. One participant models adversarial fills — longs at the next bar's high ask, shorts at the lowest bid, with a penalty above 1% of that bar's volume.
- **Look-ahead leakage is the most-confessed failure.** The cheapest diagnostic offered: shift every feature forward one bar and rerun. **If the edge survives, the leak is in the *label*, not the features** — usually a rolling statistic that touches the same bars the forward return is built from.
- **Overfitting is a trade-count question, not a time question.** Fifteen or twenty forward trades tells you nothing; a broken system passes twenty trades regularly. Two named curve-fit tells: high parameter sensitivity under 5–10% input noise, and a per-trade distribution where the mean is good but the *median* is not.
- **Sizing, not strategy, is the reported killer.** "Most people diagnose a bad exit as a strategy problem when it was actually a position size that made clear thinking impossible."

## 12.2 Repositories worth opening, and the flood to ignore

Two are worth actually opening: a C++ options market-making engine that puts SVI *inside an execution loop* rather than in a plot, and a Rust vol-surface library if you ever want the surface off the Python hot path. The mainstream backtesting and crypto plumbing repositories are covered in Section 6.

**The pattern to recognise**

A GitHub search sorted by recency returns page after page of freshly pushed, **zero-star** NIFTY and BankNifty repositories, nearly all within the last 48 hours, *none* with a stated cost model. This is the most crowded and lowest quality corner of the entire sweep. Separately, an active shill pattern is visible: multiple new gamma-exposure dashboards launched in the same 30-day window, and one participant's summary — "everyone without an edge has pivoted to using AI coding assistants to build and sell a service." **Treat every new gamma dashboard in this window as marketing until proven otherwise.**

## 12.3 Honest negatives about this sweep

- **Hacker News is dead on this topic** — nothing above single-digit points in the window. Drop it from future sweeps.
- **The careers-focused quant subreddit is pure noise** — interview prep and "how do I break in". Zero technical content. Drop it.
- **Crypto quant chatter did not appear on Reddit at all.** Given your 40% crypto weighting this is a **coverage failure, not a finding** — crypto vol discussion lives on X, Discord and Telegram, and X was not swept because no credentials were available. **Re-run the crypto leg with authenticated access before drawing any conclusion from its absence.**
- **No cost model was stated anywhere in this chatter layer.** Under your own filter almost nothing here would survive to a recommendation — which is the correct outcome for a chatter agent. This is a routing list, not a reading list.

## 13. What nobody has published: your openings

Every item below is something round one or round two searched for and could not find, on a question you are unusually well-positioned to answer. They are ordered by expected value per week of effort.

1. **Where does an SVI fit residual exceed its own error bar on NIFTY weeklies, and does it mean-revert net of the tick and STT?** Specified as a pre-registered falsification test in Section 4.3. **A rigorous public study of this would be the first of its kind that could be found.** It is a paper and a strategy at the same time, and the honest expectation is that it fails — which is itself the result.
2. **The deseasonalised realised-variance versus remaining-implied-variance study on NIFTY minute data**, dollar-gamma weighted, with the jump decomposition reported alongside. Specified step by step in Section 5.2.5. Invisible in end-of-day data, and it yields the active/inactive overnight variance ratio for NIFTY — a publishable number nobody has estimated.
3. **A public, runnable Indian SPAN implementation** driven by NSE's own free daily files. No such thing exists; the parameter file is member-gated but everything that matters for index derivatives is a published constant or a public file. **About sixty lines of Python**, and a better use of a week than reading another paper.
4. **The asymmetric hedging band for a one-sided cost.** STT is sell-side only, which violates the symmetric-cost assumption underlying Leland, Whalley–Wilmott, Barles–Soner and Zakamouline alike. The true no-trade region is asymmetric and nobody has published this for the Indian market. **Days, not weeks.**
5. **Hull–White minimum-variance delta coefficients for NIFTY**, estimated on your own minute data and *conditioned on position in the weekly Tuesday cycle* — a structure the US literature does not have. Nobody has published these, and the original work explicitly excluded sub-14-day options, which is where all your volume is.
6. **Business-time theta calibration for NIFTY weeklies.** With a Tuesday expiry the weekend sits in the *middle* of the option's life, and Indian holidays cluster unevenly, so a weekly spans seven calendar days but three to five trading days. Nobody has published on this, and minute data closes it.
7. **Crypto variance risk premium measured with a real cost model, 2024–2026.** The rigorous academic measurement stops earlier and includes no costs; everything after it is vendor assertion. Free multi-year implied-vol index history plus obtainable spot settles it. **A two-day project and genuinely original.**
8. **Dealer positioning on NIFTY, with the sign estimated rather than assumed**, validated against the free daily participant-category file as an aggregate constraint. No NIFTY study exists; the academic literature is entirely S&P 500.
9. **The weekend theta trade in crypto, costed.** The venue uses pure calendar time by published convention, so the Friday implied-vol dip is real and observable. Whether it survives a 25%-of-premium round trip on short-dated options is unpublished.
10. **Delta-neutral hedging bands with a funding carry term.** Perpetual funding introduces a carry term in the hedge leg that no classical band literature models, and it can dominate the transaction-cost term. Unfilled.
11. **A time series of the Deribit-versus-domestic-venue liquidity comparison.** Section 8 gives the method and a single snapshot. Nobody has published the series, and you can record it.

**The common thread**

Nine of these eleven require exactly two things you already have or are about to have: **minute-resolution data and a working surface fitter**. None requires capital. Most take days to weeks. And several are simultaneously a piece of original research and a falsification test of something you would otherwise have believed — which is the cheapest possible way to learn a market.

# 14. Appendices

## 14.1 Paid, worth considering later

Listed separately and never mixed into the recommendations. Check institutional access before buying any of it — see the access note in Section 10.

**Books, in priority order.**

1. **Sinclair, *Volatility Trading* (2nd ed.)** — the one book with no free substitute. Position sizing under non-normal option returns, the real economics of the variance premium, and an unusually long treatment of why most published option edges are cost artefacts.
2. **Bergomi, *Stochastic Volatility Modeling*** — but read the two free chapters and the free *Smile Dynamics* papers first; they carry perhaps 40% of the practical value at zero cost.
3. **Taleb, *Dynamic Hedging*** — for the taxonomy of risk and the only book-length treatment of vega bucketing by someone who did it on a desk.
4. **Bouchaud et al., *Trades, Quotes and Prices*** — but the free 100-page precursor by the same authors covers a large fraction of it.
5. **Sinclair, *Positional Option Trading*** — second, not first; substantial overlap with the above.
6. **Aït-Sahalia & Jacod, *High-Frequency Financial Econometrics*** — the reference behind the realised-volatility section. Library, not purchase.

**Two paywalled papers worth pulling through your institution:** the Dynamic Nelson–Siegel smile-harvesting paper on NSE data (**check whether it nets transaction costs — this is the single most important open question in the Indian material**); and the paper establishing the Indian single-stock implied-over-realised premium that governs the dispersion sign.

**Data, not education, and only after your pipeline is proven on free samples:** continuous tick-level crypto options history; institutional crypto derivatives feeds — likely reachable through your institution before paying retail.

**Explicitly not recommended at any price:** the proprietary-firm options programmes, the mentoring businesses, and the entire Indian option-selling course market.

## 14.2 Primary documents — the index

The documents to read yourself rather than take on trust. All free.

**Indian regulation and market structure**

- SEBI, *Measures to Strengthen Equity Index Derivatives Framework*, 1 Oct 2024 — the six measures and their effective dates.
- SEBI, *Final Settlement Day (Expiry Day) for Equity Derivatives Contracts*, 26 May 2025 — and the NSE implementation circulars, which are where the backtest trap lives.
- NSE circular on lot-size revision, 3 Oct 2025.
- SEBI, *Review of eligibility criteria for entry/exit of stocks in derivatives segment*, 30 Aug 2024.
- SEBI Master Circular for Stock Exchanges and Clearing Corporations, **Chapter 5** — the complete SPAN specification, the 16-scenario table, ELM rates, and the peak-margin framework.
- NSE Clearing risk-management FAQ — SPAN file cadence, the ELM ladder, and the beginning-of-day parameter rule that makes margin simulation feasible.
- NSE Clearing settlement-price and STT pages — the method, and the per-client STT report file spec you can parse.
- SEBI, *Profitability* and *Trading Behaviour of Individual Traders in the Equity Derivatives Segment (FY25–FY26)*, Aug 2026 — the current numbers, superseding everything earlier.

**Indian daily data files, all anonymously downloadable**

- Market lot file — current lot size per underlying per expiry month.
- Daily volatility file — EWMA daily and annualised volatility per symbol, **the exact input that feeds the price and volatility scan ranges**.
- F&O bhavcopy — OHLC, settlement price, open interest, lot size, underlying price, per contract.
- Settlement price file — the number determining exercise STT intrinsic value.
- Collateral haircut files.
- **Participant-wise open interest, signed by category, daily** — the file in Section 4.4 with no US equivalent.
- The circulars JSON index — every circular with a direct PDF link.

These require a cookie bootstrap and a browser user-agent; nothing else.

### Tax

- Income-tax Act 2025, official gazette text — ss. 66(31), 66(33), 108–113, 121 for the F&O treatment; s.194(1) Table Sl. 4 and s.393(1) Table Sl. 8(vi) for virtual digital assets.
- Finance Bill/Act 2026 — the operative STT amendment clause.
- NSE STT rate table and tax circular — the definitive levy table.
- ICAI Guidance Note on tax audit — the only authoritative statement of how derivative turnover is computed.
- CBDT ITR-2 validation rules — the schedule that enforces transaction-level VDA reporting.

### Foreign exchange

- RBI Master Direction on the Liberalised Remittance Scheme.
- **FEM (Overseas Investment) Directions 2022, definition (ix)(a)(iii)** — the single clause that decides the US-venue question, and the one every secondary source paraphrases wrongly.

### Crypto venue specifications

- Exchange API reference and contract introduction policy — the listing ladder and strike intervals.
- Index-price and settlement pages — the capped-median index and the 30-minute settlement window, including the fact that expiring-option delta decays linearly to zero over the final thirty minutes, which will silently break a naive hedger.
- Portfolio margin model — a real specification with formulas. **Note the published deck is superseded**; use it for structure and pull live parameters from your own account.
- Inverse-options specification — the exchange's exact Black-Scholes convention, including that implied vol uses the *forward*, and that time is pure calendar time.
- Fee schedule — the 12.5%-of-premium cap.

### Method and measurement

- BCBS d457, *Minimum capital requirements for market risk* — MAR32 for the P&L attribution thresholds, MAR21 for the published vega-bucketing scheme.
- The 0DTE replication package and its `KNOWN-ISSUES` file — the best-documented cost-scale failure in the public options literature.

## 14.3 What could not be verified

Collected in one place, because a survey that hides its gaps is worse than one that shows them.

- **Whether the DNS smile paper nets transaction costs** — paywalled, no preprint found. The single most important open question in the Indian material.
- **SEBI turnover fee and stamp-duty rates** on derivatives; **GST at 18%** on brokerage. Together about 0.3 bp; they move no conclusion.
- **Whether NSE transaction charges are still at the September 2024 level.**
- **Whether the November 2024 expiry-day ELM add-on has since been modified** by a later risk circular.
- **The consolidated "as amended" Income-tax Act PDF** — 403-blocked; the gap was closed by reading the amendment clauses directly.
- **Whether the 1% withholding attaches to VDA *derivatives*** — genuinely unresolved in the statutory text. A gap in the law, not a research failure.
- **The client-level 50% cash-margin requirement** — verified at clearing-member level only.
- **A public register of registered crypto service providers** — none exists, so a venue's stated registration is a self-declaration.
- **Current numeric portfolio-margin parameters** on the crypto venue — exposed only to logged-in accounts.
- **The Sinclair sizing paper's body** — the repository blocks programmatic access, so the abstract was read but not the paper, and no formula from it is reproduced. It is free and short; download it.
- **Several classical hedging papers** — Whalley & Wilmott (1997), Hodges & Neuberger (1989), Davis–Panas–Zariphopoulou (1993) — were not reached in the original. A free working paper restates all three faithfully and was used as a labelled substitute.
- **Gaussian HMMs failing to reproduce volatility clustering** — stated in the jump-model literature, not independently confirmed. Cheap to check by simulation.
- **Chapter-level content for several books** is from prior knowledge rather than from opening them

this round. Chapter *titles* should be treated as approximately, not exactly, correct.
- **X and YouTube were not swept** in the social layer. A named, acknowledged hole.

**One closing note on method**

The most valuable single output of this exercise was not a source. It was the discovery that three claims round one presented confidently — that the US was reachable, that the standard dealer-positioning construction applies to NIFTY, and that the SEBI studies show retail is short volatility — were all **false**, and that a fourth, the SVI residual, was **unresolvable on the data round one used**. Every one of those was caught by making a second pass whose only job was to attack the first. If you take one process habit from this document, take that one.